

ĐẠI HỌC QUỐC GIA HÀ NỘI  
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



Le Xuan An

Gradient-Routed Alignment with Capacity-aware Experts  
for Multimodal Understanding

KHOÁ LUẬN TỐT NGHIỆP

Ngành: Khoa học máy tính

Giảng viên hướng dẫn: Tiến sĩ Nguyễn Văn Sơn

Giảng viên đồng hướng dẫn: Phó Giáo sư Võ Đình Hiếu

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VIETNAM NATIONAL UNIVERSITY, HANOI  
UNIVERSITY OF ENGINEERING AND TECHNOLOGY



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**Gradient-Routed Alignment with Capacity-aware Experts  
for Multimodal Understanding**

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**Advisor: Dr. Nguyen Van Son**

**Co-Supervisor: Associate Professor Vo Dinh Hieu**

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## Lời cảm ơn

Lời đầu tiên, tôi xin gửi lời cảm ơn chân thành và lòng biết ơn sâu sắc tới thầy TS. Nguyễn Văn Sơn và thầy PGS.TS. Võ Đình Hiếu vì đã hướng dẫn và hỗ trợ tôi về kiến thức chuyên môn vô cùng tận tình trong quá trình nghiên cứu và thực hiện khóa luận này.

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Cuối cùng, tôi xin gửi lời cảm ơn chân thành tới các thầy, cô trong trường Đại học Công Nghệ - Đại học Quốc gia Hà Nội đã tạo điều kiện cho tôi được học tập và nghiên cứu để có thể hoàn thành khóa luận với kết quả tốt nhất.

## **Assurance**

I hereby affirm that this thesis is my own original work. I have not submitted it previously for any academic or professional award.

I completed this thesis through my own efforts, with guidance and support from Dr. Nguyen Van Son and Associate Professor Vo Dinh Hieu, in partial fulfillment of the graduation requirements. I confirm that all information, data and findings presented here are original and free from plagiarism.

I have acknowledged every source that I consulted and included each one in the references. I accept full responsibility for the accuracy and integrity of this thesis under the regulations of VNU University of Engineering and Technology.

Hanoi, Ngày 15 tháng 5 năm 2026

Student

Le Xuan An

## Tóm tắt

Khóa luận này nghiên cứu bài toán chứng cất tri thức đa giáo viên không đồng nhất cho mô hình thị giác-ngôn ngữ (VLM). Mục tiêu là truyền tri thức từ một tập mô hình giáo viên đồng bằng sang một mô hình học sinh nhỏ hơn khi các giáo viên thuộc những họ mô hình khác nhau, sử dụng bộ tách từ khác nhau và có mức độ hữu ích thay đổi theo từng mẫu. Trong bối cảnh này, chứng cất trực tiếp theo chỉ số token không còn hợp lệ và phép lấy trung bình cố định giữa giáo viên có thể trộn tín hiệu hữu ích với tín hiệu không liên quan hoặc xung đột.

Để giải quyết bài toán trên, khóa luận đề xuất Gradient-Routed Alignment with Capacity-aware Experts (GRACE), một khung chứng cất cho truyền tri thức VLM đa giáo viên không đồng nhất. GRACE kết hợp ba cơ chế: chứng cất trie-Wasserstein trên cây byte UTF-8 dùng chung, định tuyến giáo viên thừa top- $k$  từ trạng thái ẩn của mô hình học sinh và tinh chỉnh trọng số bằng độ đồng thuận gradient. Hàm mất mát trie-Wasserstein giúp so sánh phân phối đầu ra của giáo viên và học sinh giữa các bộ tách từ khác nhau. Định tuyến thừa chọn các giáo viên phù hợp với từng mẫu. Tinh chỉnh bằng độ đồng thuận gradient sau đó điều chỉnh trọng số đã định tuyến theo mức độ căn chỉnh giữa gradient chứng cất của từng giáo viên và gradient cross-entropy có giám sát. Thực nghiệm trên TextVQA, DocVQA và ChartQA đánh giá GRACE với hai mô hình học sinh SmolVLM-500M và SmolVLM-256M trong thiết lập bốn giáo viên không đồng nhất. Với SmolVLM-500M, GRACE đạt 69.38, 73.11 và 80.20 trên TextVQA, DocVQA và ChartQA, so với điểm gốc 56.75, 59.52 và 62.80 của mô hình học sinh. Các kết quả này tương ứng với mức cải thiện tương đối 22.26%, 22.83% và 27.71% so với mô hình gốc, đồng thời thu hẹp 55.03%, 39.53% và 55.34% khoảng cách ban đầu tới giáo viên có điểm cao nhất. Với SmolVLM-256M, GRACE đạt 60.73, 60.48 và 72.32 trên cùng ba bộ đánh giá. Trên các thực nghiệm được báo cáo, GRACE vượt qua tinh chỉnh trực tiếp, chứng cất logits cùng họ mô hình, ULD khác họ mô hình và các phương pháp đa giáo viên không đồng nhất được so sánh.

**Từ khóa:** mô hình thị giác-ngôn ngữ, chứng cất tri thức, chứng cất đa giáo viên, định tuyến giáo viên thừa, đồng thuận gradient, chứng cất Wasserstein khác bộ tách từ

# Abstract

This thesis studies heterogeneous multi-teacher knowledge distillation for vision-language models (VLMs). The problem is to transfer supervision from a pool of frozen teacher models to a smaller student when the teachers come from different model families, use different tokenizers and vary in usefulness across samples. In this setting, direct token-index distillation is invalid and fixed teacher averaging can mix helpful teacher signals with irrelevant or conflicting ones.

To address this problem, the thesis proposes Gradient-Routed Alignment with Capacity-aware Experts (GRACE), a distillation framework for heterogeneous teacher-to-student VLM transfer. GRACE combines three mechanisms: trie-Wasserstein knowledge distillation over a shared UTF-8 byte trie, sparse top- $k$  teacher routing from student hidden states and gradient-agreement refinement of routed teacher weights. The trie-Wasserstein loss makes teacher and student output distributions comparable across tokenizers. Sparse routing selects sample-relevant teachers. Gradient-agreement refinement then adjusts routed weights according to whether each teacher’s distillation gradient aligns with the supervised cross-entropy gradient.

Experiments on TextVQA, DocVQA and ChartQA evaluate GRACE with SmolVLM-500M and SmolVLM-256M students under a four-teacher heterogeneous pool. For SmolVLM-500M, GRACE reaches 69.38, 73.11 and 80.20 on TextVQA, DocVQA and ChartQA, compared with original student scores of 56.75, 59.52 and 62.80. These results correspond to relative gains of 22.26%, 22.83% and 27.71% over the original student and close 55.03%, 39.53% and 55.34% of the original gap to the highest-scoring teacher. For SmolVLM-256M, GRACE reaches 60.73, 60.48 and 72.32 on the same benchmarks. Across the reported experiments, GRACE improves over direct finetuning, same-family logits KD, cross-family ULD and the compared heterogeneous multi-teacher baselines.

**Keywords:** vision-language model, knowledge distillation, multi-teacher distillation, sparse teacher routing, gradient agreement, cross-tokenizer Wasserstein distillation

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# List of Abbreviations

| Abbr.   | Full Term                                             | Description                                                                |
|---------|-------------------------------------------------------|----------------------------------------------------------------------------|
| ANLS    | Average Normalized Levenshtein Similarity             | Evaluation metric used for DocVQA.                                         |
| CE      | Cross-Entropy                                         | Supervised answer-token loss used during training.                         |
| CKA     | Centered Kernel Alignment                             | Representation similarity measure used to compare hidden layers.           |
| ChartQA | Chart Question Answering                              | Benchmark for question answering over charts.                              |
| DocVQA  | Document Visual Question Answering                    | Benchmark for question answering on document images.                       |
| EOS     | End of Sequence                                       | Special token that marks the end of a generated answer span.               |
| GRACE   | Gradient-Routed Alignment with Capacity-aware Experts | Proposed multi-teacher distillation method in this thesis.                 |
| SHM-CKA | Hidden-State Matching with CKA                        | Baseline that combines logits distillation with hidden-state CKA matching. |
| JS      | Jensen-Shannon                                        | Symmetric divergence measure between probability distributions.            |
| KD      | Knowledge Distillation                                | Transfer of supervision from teacher models to a student model.            |
| KL      | Kullback-Leibler                                      | Divergence used to compare probability distributions.                      |
| LLM     | Large Language Model                                  | Language model with large parameter scale and generation capability.       |
| MLP     | Multi-Layer Perceptron                                | Feedforward projection block used in the teacher routing network.          |
| MoE     | Mixture of Experts                                    | Sparse routing framework that motivates teacher selection.                 |
| MTKD    | Multi-Teacher Knowledge Distillation                  | Distillation setting that uses more than one teacher model.                |

| Abbr.        | Full Term                            | Description                                                                          |
|--------------|--------------------------------------|--------------------------------------------------------------------------------------|
| OCR          | Optical Character Recognition        | Extraction of text from images or scanned documents.                                 |
| OT           | Optimal Transport                    | Distribution-matching view used to motivate trie-Wasserstein KD.                     |
| TWD          | Tree-Wasserstein Distance            | Tree-structured optimal-transport distance used as background for the proposed loss. |
| TextVQA      | Text Visual Question Answering       | Benchmark for question answering on natural images containing text.                  |
| ULD          | Universal Logit Distillation         | Tokenizer-agnostic distillation baseline based on ranked probability distributions.  |
| VLM          | Vision-Language Model                | Model that processes visual and textual inputs together.                             |
| VLM-Eval-Kit | Vision-Language Model Evaluation Kit | Toolkit used to run standardized multimodal evaluation.                              |
| VQA          | Visual Question Answering            | Task of answering questions grounded in visual inputs.                               |

# Chapter 1

## Introduction

**Vision-language models (VLMs)** have emerged as a **central direction** in modern AI because they integrate visual perception with language reasoning. By coupling powerful image encoders with autoregressive language decoders through learned multimodal connectors, these models can interpret visual content, read text embedded in images and reason across modalities in ways that neither vision systems nor language models alone can easily achieve [1–3]. This architectural synthesis has unlocked a **broad set of capabilities**, including extracting information from scanned documents, understanding visually rich scenes and grounding language instructions in physical environments [4–8].

Driven by neural scaling laws, the most capable VLMs are usually built on large language backbones trained with large multimodal corpora. This recipe reliably improves performance but also raises computational cost [9, 10]. A recent open VLM family shows the same spread: Qwen2-VL studies 2B, 7B and 72B variants, so the largest reported variant is **36 times** larger than the smallest one [11]. As model size grows, the gap between what is achievable in research and what is deployable in practice widens accordingly. The **highest-scoring models are often the hardest to deploy** when latency, memory footprint, energy consumption and hardware constraints matter. In robotics and embodied AI, an agent must process visual observations, interpret language instructions and produce action decisions in real time, often on resource-constrained hardware [7, 8]. Similar constraints appear in mobile applications, document-processing pipelines and enterprise deployments, where users and operators need models with low latency, controlled cost and clear security boundaries. This widening **gap between frontier capability and deployment constraints** creates a central tension for applied VLM research.

**Knowledge distillation** offers a principled path through this tension by compressing the behavior of a large teacher model into a compact student. The student avoids training from scratch on the same large-scale corpora and learns from the teacher’s output distributions and internal representations, which carry richer information than hard labels alone [12, 13]. Distillation has been widely used to produce smaller models that **retain much of the capability of their larger counterparts**, and recent lightweight VLMs demonstrate that this direction is viable for latency-sensitive and cost-sensitive deployment [14–16]. Existing approaches each address part of the challenge, but **none covers it fully**. Response-based distillation aligns teacher and student output distributions effectively when both share a tokenizer, but **breaks down** when teacher and student use **different**

**vocabularies** [12, 13]. Transport-based methods relax this assumption by comparing probability mass over decoded strings, but they are usually formulated for single-teacher transfer and do not determine which teacher should supervise a given sample [17–19]. Representation matching through hidden states or similarity measures such as CKA provides a tokenizer-independent signal, but does not directly optimize answer likelihood and **cannot resolve conflicts between teachers** [20]. Multi-teacher distillation methods show the value of pooling several teachers, but fixed or globally learned weights can mix useful supervision with low-utility or conflicting signals [21–23].

These limitations are **most severe** in text-rich visual reasoning tasks such as document question answering, chart interpretation and scene text understanding. Although recent open-source VLMs have narrowed the gap with proprietary systems on common benchmarks [11, 24, 25], smaller models continue to struggle when fine visual details, document structure and numerical relations must be captured. The central motivation of this thesis is to combine **tokenizer-agnostic** distribution matching, **sample-dependent** teacher selection and **gradient-aware conflict resolution** for heterogeneous VLM distillation.

This thesis evaluates **Gradient-Routed Alignment with Capacity-aware Experts (GRACE)** using VLM-Eval-Kit on TextVQA, DocVQA and ChartQA with SmolVLM-500M and SmolVLM-256M as student models. In our setting, GRACE moves SmolVLM-500M closer to the highest-scoring teacher by closing 55.03%, 39.53% and 55.34% of original teacher-student gaps of 22.95, 34.38 and 31.44 points on TextVQA, DocVQA and ChartQA, supporting deployment of student VLMs in VQA scenarios. The research questions focus on performance, component effects and training cost:

- **RQ1: Performance.** How does GRACE compare with direct fine-tuning, same-family logits KD, ULD and heterogeneous multi-teacher baselines on TextVQA, DocVQA and ChartQA benchmarks?
- **RQ2: Component Analysis.** How much do trie-Wasserstein KD, sparse top- $k$  routing and gradient-agreement refinement contribute to the final performance?
- **RQ3: Efficiency.** What execution-time and computational overhead does GRACE introduce during training?

This thesis addresses this gap through three contributions. First, a **trie-Wasserstein KD loss** maps teacher and student vocabularies onto a shared UTF-8 byte trie and computes a  $W_1$  optimal-transport distance through subtree-mass imbalances. Unlike standard ULD losses, this objective **preserves the lexical structure** of decoded token strings: tokens that share a long byte prefix are cheaper to transport across than unrelated ones, making the loss well-suited to OCR-heavy outputs where correct and near-correct answers often differ only in prefixes, suffixes or whitespace conventions. Second, **gradient-agreement refinement** scores each routed teacher by comparing its KD gradient against the supervised CE gradient using cosine similarity, stabilizes the scores with an exponential moving average and geometrically blends them with the router’s own weights, ensuring that only teachers whose update directions are **compatible** with the supervised objective **can contribute** to training. Finally, **GRACE** integrates the trie-Wasserstein loss, a sparse top- $k$  teacher routing network and gradient-agreement refinement into a single end-to-end objective, where each component **corrects one failure the previous cannot fix**. Evaluated on TextVQA, DocVQA and ChartQA with SmolVLM-500M and SmolVLM-256M under a four-teacher pool, GRACE improves over direct fine-tuning, same-family KD, ULD and heterogeneous multi-teacher baselines, closing **up to 55% of the teacher-student gap**.

The remainder of this thesis is structured as follows. Chapter 2 provides background on VLMs, knowledge distillation, tokenizer-mismatched distribution matching, sparse routing and gradient conflict resolution. Chapter 3 presents GRACE in full, covering the trie-Wasserstein objective, sparse teacher router, gradient-agreement refinement and combined training objective. Chapter 4 describes the experimental setup, datasets and evaluation metrics, baselines and ablation settings. Chapter 5 reports the main comparison, ablation study, efficiency analysis and discussion. The thesis concludes with the main findings and future directions.

# Chapter 2

## Background and Related Work

This chapter introduces the background concepts and related work used throughout the thesis. It first covers the building blocks of VLMs and knowledge distillation, then turns to the cross-tokenizer, routing and multi-teacher challenges. The final section connects these threads into the research gap addressed by this thesis.

### 2.1 Vision-Language Models

Modern vision-language models often combine a vision encoder with a language model. On the visual side, many systems use a Vision Transformer pretrained with an image-text objective, commonly CLIP [26] or SigLIP [27], which converts an image into a sequence of patch embeddings that capture both local and global visual features. A connector module then fuses these visual tokens with text, using cross-attention, a Q-Former or a learned projector [1–3]. This design separates visual perception from language generation: the vision encoder produces a compact sequence of image features, while the language model provides the autoregressive prior used to answer questions. A useful way to distinguish VLM families is by where and how visual-language fusion happens, as shown in Figure 2.1. Dual-encoder models align image and text embeddings contrastively, cross-attention models let language tokens attend to visual features directly and projector-based models map visual embeddings into the language model hidden space and then decode with a standard causal language model. This thesis focuses on the projector-based generative design, because distillation targets output token distributions, not retrieval embeddings. In this setting, projected image embeddings are inserted into the language-model context as visual tokens, so answer generation remains ordinary next-token prediction conditioned on both the image representation and the textual question.

---

<sup>2</sup>Panels adapted from CLIP [26], Flamingo [1], BLIP-2 [2] and LLaVA [3].

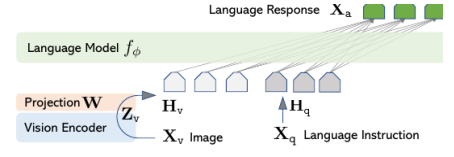
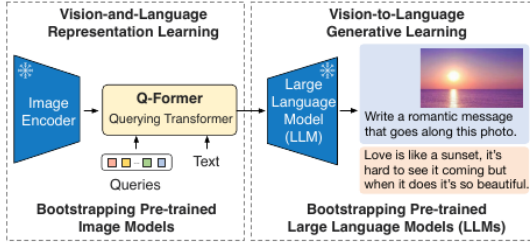
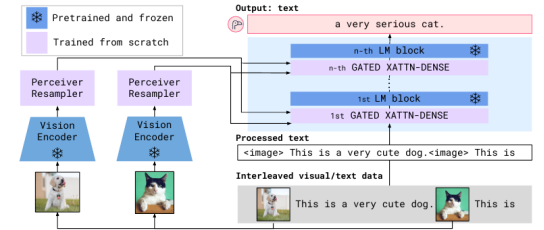
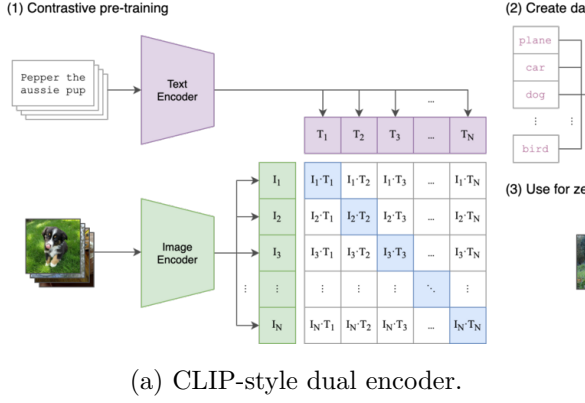


Figure 2.1: Representative architecture families of modern vision-language models.<sup>2</sup>

For the generative setting studied in this thesis, the student VLM produces answer tokens autoregressively. For language modeling, the Transformer maps each hidden state  $\mathbf{h}_j$  to next-token logits and probabilities [28]:

$$\mathbf{z}_j = \mathbf{W}_o \mathbf{h}_j + \mathbf{b}_o, \quad p_j(v) = \text{softmax}(\mathbf{z}_j)_v = \frac{\exp(z_{j,v})}{\sum_{u \in \mathcal{V}} \exp(z_{j,u})}. \quad (2.1)$$

At answer position  $j$ , the VLM can be abstracted as a mapping from the multimodal input and previous answer tokens to next-token logits:

$$\mathbf{z}_j = f_\theta(x, y_{<j}), \quad \mathbf{z}_j \in \mathbb{R}^{|\mathcal{V}|},$$

where  $x$  is a multimodal input and  $\mathbf{z}_j$  denotes the logits used for autoregressive answer generation at position  $j$ . For an answer sequence  $y = (y_1, \dots, y_L)$ , the model factorizes the conditional probability as

$$p_\theta(y \mid x) = \prod_{j=1}^L p_\theta(y_j \mid x, y_{<j}). \quad (2.2)$$

Token-level distillation therefore compares teacher and student answer-token distributions at supervised positions, such as  $p_{t,j}$  and  $p_{s,j}$  [12, 28]. This is important for VQA because short answers such as numbers, names and document fields may differ by only one or two tokens, so supervision must preserve position-level information, not only sequence-level

accuracy after decoding the full answer. This output-centric view motivates the next sections: standard KD compares teacher and student answer-token distributions directly, while heterogeneous VLM distillation needs a tokenizer-agnostic matching objective.

## 2.2 Knowledge Distillation

Knowledge distillation transfers information from a higher-capacity teacher model to a student model [12, 13]. Let  $\mathcal{L}_{CE}$  denote the supervised cross-entropy loss of the student. A generic distillation objective takes the form

$$\mathcal{L} = \lambda_{CE}\mathcal{L}_{CE} + \lambda_{KD}\mathcal{L}_{KD}, \quad (2.3)$$

where  $\mathcal{L}_{KD}$  is a teacher-guided loss. In token-level distillation, the teacher and student are compared on their output distributions, not only on hard labels. The student therefore learns richer distributional structure: the relative probabilities over alternative classes, often called *dark knowledge*, which hard labels discard [12]. For a single answer position, a common same-vocabulary form is

$$\mathcal{L}_{KD} = \tau^2 \text{KL}\left(\mathbf{p}_t^{(\tau)} \parallel \mathbf{p}_s^{(\tau)}\right), \quad (2.4)$$

where  $\tau$  is the distillation temperature. The cross-entropy term anchors the student to the ground-truth answer, while the KD term transfers the teacher’s relative preference over alternative tokens. In the heterogeneous setting of this thesis, this template remains useful, but the comparison term must be changed because teachers and students may not share the same tokenizer.

The output distributions used in KD are often softened before comparison. Let  $\mathbf{z}_s$  and  $\mathbf{z}_t$  denote student and teacher logits. With temperatures  $\tau_s$  and  $\tau_t$ , the corresponding distributions are

$$\mathbf{p}_s = \text{softmax}\left(\frac{\mathbf{z}_s}{\tau_s}\right), \quad \mathbf{p}_t = \text{softmax}\left(\frac{\mathbf{z}_t}{\tau_t}\right). \quad (2.5)$$

Many classical KD methods compare  $\mathbf{p}_s$  and  $\mathbf{p}_t$  using divergences such as forward KL or Jensen-Shannon divergence. However, such approaches usually assume that teacher and student share the same vocabulary [12, 17, 29]. The temperature controls how much probability mass is exposed beyond the top answer token. When  $\tau > 1$ , the softmax distribution becomes flatter and low-rank alternatives receive non-negligible probability. When  $\tau < 1$ , the distribution becomes sharper and approaches hard-label supervision. This matters because the teacher may encode useful semantic alternatives in the tail of the distribution, for example related words, number formats or subword variants. Section 3.2 applies the same softening idea before trie-Wasserstein tokenizer-independent

distribution matching. Since student and teacher temperatures may be chosen independently,  $\tau_t$  controls how much dark knowledge is exposed from the teacher, whereas  $\tau_s$  controls the smoothness of the student distribution being matched. For this reason, the temperatures are treated as part of the distillation interface throughout this thesis, not merely numerical constants inside the softmax.

For multimodal question answering, distillation should focus on *answer tokens*, not *prompt tokens* [3]. The KD loss is therefore applied only on supervised answer positions after masking ignored tokens and truncating student and teacher answer spans to a common length. The result is a token-aligned distillation signal that remains compatible with different teacher and student tokenizers. Let  $\Omega_b^{CE}$  denote the non-ignored student answer positions for sample  $b$ , so the supervised loss is

$$\mathcal{L}_{CE}^{(b)} = \frac{1}{|\Omega_b^{CE}|} \sum_{j \in \Omega_b^{CE}} -\log p_s(y_j \mid x_b, y_{<j}). \quad (2.6)$$

The same masking principle is used for KD positions. This prevents the distillation objective from spending capacity on prompt reconstruction, padding tokens or formatting artifacts and concentrates the teacher signal on the answer span that is scored by the evaluation benchmarks. For multiple teachers, the KD position set can be teacher-specific because teacher tokenizers and generated answer spans may differ. The training code therefore constructs  $\Omega_{b,i}^{KD}$  separately for each teacher  $T_i$ , then averages only over valid aligned answer positions for that teacher.

Knowledge distillation methods can also be grouped by the type of teacher signal they transfer. Common response-level divergences and feature-level similarity losses used in knowledge distillation are collected in Table 2.1 [12, 29]. In the table,  $p$  and  $q$  denote teacher and student output distributions over token events  $t$ .  $m = \frac{1}{2}(p + q)$  is the JS midpoint distribution.  $\mathbf{u} = \Phi_T(f_T(x, y))$  and  $\mathbf{v} = \Phi_S(f_S(x, y))$  are teacher and student feature vectors. The index  $r$  marks feature dimensions or normalized feature bins. *Response-based* distillation uses the teacher’s output distribution or generated response as supervision. In its standard form, the loss compares next-token distributions:

$$\mathcal{L}_{resp} = \frac{1}{|\Omega|} \sum_{j \in \Omega} \text{KL}(\mathbf{p}_{t,j}^{(\tau)} \parallel \mathbf{p}_{s,j}^{(\tau)}), \quad (2.7)$$

where  $\Omega$  is the set of supervised answer positions. This objective is closely tied to the student’s final prediction layer and therefore provides a task-facing signal. It is the classical form of KD introduced by Hinton et al. [12] and remains common in language-model and VLM compression [13, 30].

*Feature-based* distillation aligns intermediate representations between student and teacher layers. Given student and teacher hidden states  $\mathbf{H}_s^{(\ell)}$  and  $\mathbf{H}_t^{(m)}$  at selected layers, a

direct matching objective is

$$\mathcal{L}_{feat} = \left\| \phi_s(\mathbf{H}_s^{(\ell)}) - \phi_t(\mathbf{H}_t^{(m)}) \right\|_2^2, \quad (2.8)$$

where  $\phi_s$  and  $\phi_t$  are projection functions that make the hidden states comparable. Feature-based methods are useful when output vocabularies are incompatible, because the comparison can be defined in representation space without vocabulary-space matching [20, 31]. However, they are less directly connected to answer likelihood and may transfer teacher-specific internal structure that is not useful for the downstream task. This distinction motivates treating response-level and representation-level signals as complementary, not interchangeable [31, 32].

Table 2.1: Summary of response-level and feature-level KD objectives.

| Response-Level Divergences |                                                             | Feature-Level Losses     |                                          |
|----------------------------|-------------------------------------------------------------|--------------------------|------------------------------------------|
| Type                       | $D(p, q)$                                                   | Function $\mathcal{L}_F$ | Expression                               |
| Forward KLD                | $\sum_t p(t) \log \frac{p(t)}{q(t)}$                        | L2-Norm Distance         | $\ \mathbf{u} - \mathbf{v}\ _2$          |
| Reverse KLD                | $\sum_t q(t) \log \frac{q(t)}{p(t)}$                        | L1-Norm Distance         | $\ \mathbf{u} - \mathbf{v}\ _1$          |
| JS Divergence              | $\frac{1}{2} \text{KL}(p\ m) + \frac{1}{2} \text{KL}(q\ m)$ | Cross-Entropy Loss       | $-\sum_r \mathbf{u}_r \log \mathbf{v}_r$ |

## 2.3 Tokenizer-Agnostic Distribution Matching

Same-vocabulary KD assumes that teacher and student probabilities are indexed by the same token set. Heterogeneous VLM distillation violates this: a teacher may tokenize the string “invoice” as one token while the student splits it into several subword pieces. Token-index KL then compares unrelated bins and is invalid [17]. The natural fix is to decode each token into a string and compare probability mass over decoded strings, not vocabulary indices. Let  $\mathcal{V}_t$  and  $\mathcal{V}_s$  be the teacher and student vocabularies with decoding maps  $\psi_t : \mathcal{V}_t \rightarrow \mathcal{S}$  and  $\psi_s : \mathcal{V}_s \rightarrow \mathcal{S}$  into a shared string space  $\mathcal{S}$ . Distribution matching is then defined over decoded strings, not vocabulary indices. This motivates transport-based objectives that replace same-vocabulary KL with a distance over probability mass on a shared lexical support.

Optimal transport compares two probability measures by the minimum cost required to move mass from one to the other [33]. For probability measures  $\mu_s$  and

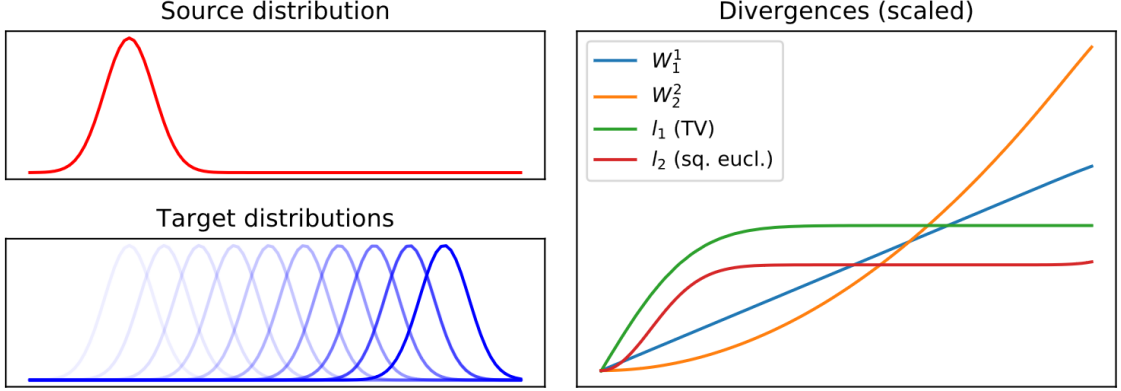


Figure 2.2: Geometric behavior of Wasserstein distance.<sup>1</sup>

$\mu_t$ , the  $p$ -Wasserstein cost is

$$W_p^p(\mu_s, \mu_t) = \min_{\gamma \in \Pi(\mu_s, \mu_t)} \int_{\Omega_s \times \Omega_t} \|x - y\|^p d\gamma(x, y), \quad (2.9)$$

where  $\Pi(\mu_s, \mu_t)$  is the set of joint couplings with marginals  $\mu_s$  and  $\mu_t$ . For discrete weights  $\mathbf{a}$ ,  $\mathbf{b}$  and cost matrix  $\mathbf{C}$ , this becomes

$$W(\mathbf{a}, \mathbf{b}) = \min_{\mathbf{P} \in \mathcal{U}(\mathbf{a}, \mathbf{b})} \langle \mathbf{P}, \mathbf{C} \rangle, \quad \mathcal{U}(\mathbf{a}, \mathbf{b}) = \{ \mathbf{P} \geq 0 \mid \mathbf{P}\mathbf{1} = \mathbf{a}, \mathbf{P}^\top \mathbf{1} = \mathbf{b} \}. \quad (2.10)$$

Unlike pointwise divergences, Wasserstein distances reflect displacement even when supports do not overlap, as Figure 2.2 illustrates.

For tokenizer-agnostic matching, dense OT is still costly because it requires a pairwise ground cost between all source and target bins. A *tree metric* provides a structured alternative: probability mass is placed on a shared hierarchy and the distance is the sum of weighted *subtree-mass imbalances* [18, 33]. For a rooted tree  $T = (V, E)$  with edge weights  $w(e)$ , one convenient form is

$$W_T(\mu, \nu) = \sum_{e \in E} w(e) |\mu(\Gamma(e)) - \nu(\Gamma(e))|, \quad (2.11)$$

where  $\Gamma(e)$  denotes the subtree below edge  $e$ . The key point is that, on a tree metric, transport decomposes into edgewise subtree-mass imbalance and avoids a dense source-target coupling [18, 19]. The computation is efficient because the comparison follows the tree hierarchy directly and avoids materializing a full pairwise transport matrix. Figures 2.3a and 2.3b show the two commonly used OT views: a tree-metric view for fast subtree-mass comparisons and a matrix view for discrete transport plans. Tree-sliced OT extends this idea by averaging Wasserstein distances over tree metrics and shows that tree metrics preserve the geometric view of OT while admitting fast closed-form computations [19].

<sup>1</sup>Adapted from Flamary and Courty [34].

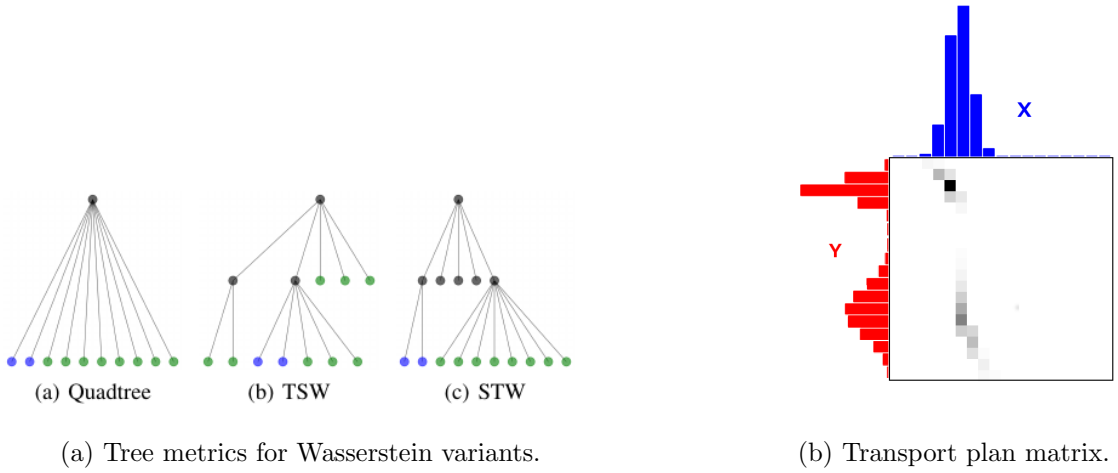


Figure 2.3: Discrete optimal transport views.<sup>2</sup>

A contrasting family of methods avoids output distributions altogether by matching intermediate representations. Representation matching transfers knowledge through hidden-state comparison, not final token distributions. Let  $\mathbf{H}_s \in \mathbb{R}^{n_s \times d_s}$  and  $\mathbf{H}_t \in \mathbb{R}^{n_t \times d_t}$  denote student and teacher hidden-state matrices. Because the sequence lengths and dimensions may differ, alignment usually requires pooling, projection, interpolation or layer selection:

$$\tilde{\mathbf{H}}_s = A_s(\mathbf{H}_s), \quad \tilde{\mathbf{H}}_t = A_t(\mathbf{H}_t), \quad \mathcal{L}_{hid} = \left\| \tilde{\mathbf{H}}_s - \tilde{\mathbf{H}}_t \right\|_F^2. \quad (2.12)$$

Here,  $A_s$  and  $A_t$  denote alignment operations that place student and teacher activations in a comparable space. This family of methods is naturally appealing for heterogeneous models because it avoids direct vocabulary comparison and can use information from image tokens, text tokens or fused multimodal states [31].

Centered Kernel Alignment (CKA) provides a complementary way to compare representations through their similarity structure, not through elementwise hidden-state equality [20]. Given two activation matrices  $\mathbf{X} \in \mathbb{R}^{n \times d_x}$  and  $\mathbf{Y} \in \mathbb{R}^{n \times d_y}$  for the same  $n$  examples, the linear CKA score can be written as

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\|\mathbf{Y}^\top \mathbf{X}\|_F^2}{\|\mathbf{X}^\top \mathbf{X}\|_F \|\mathbf{Y}^\top \mathbf{Y}\|_F}. \quad (2.13)$$

Higher CKA values indicate that the two representations induce similar geometry over the examples. Because CKA is invariant to isotropic scaling and orthogonal transformations, it is often more stable than direct hidden-state matching when the teacher and student have different hidden dimensions. In this thesis, CKA is therefore relevant both as a possible representation-level distillation signal and as an analysis tool for measuring how closely student features follow teacher features [20, 32].

<sup>2</sup>Panel (a) is adapted from Takezawa et al. [18]. Panel (b) is adapted from Wikimedia Commons and Peyré and Cuturi [33, 35].

These advantages do not make representation-only matching sufficient. Representation matching does not require a shared tokenizer, but it also does not directly optimize answer probabilities. A hidden-state objective can make the student resemble the teacher internally while still leaving the final answer distribution poorly calibrated. In a multi-teacher setting, the problem is more difficult: two teachers may have useful but incompatible internal representations and a representation-only objective does not specify which teacher should dominate for a given input. It also does not detect whether a teacher’s representation loss produces a gradient that agrees with the supervised answer loss. For these reasons, representation matching is better viewed as a complementary signal or diagnostic baseline, not a complete solution to heterogeneous multi-teacher VLM distillation [20, 31, 32].

## 2.4 Sparse Routing and Expert Selection

Recent sparse expert models route each sample or token through only a subset of available experts, improving efficiency and allowing specialized computation [36–38]. Let  $\mathbf{h}^{(b)}$  denote a pooled representation for sample  $b$ . A router network  $G_\phi(\cdot)$  produces gate logits

$$\mathbf{r}^{(b)} = G_\phi(\mathbf{h}^{(b)}), \quad \tilde{\mathbf{r}}^{(b)} = \text{AdjustedScores}(\mathbf{r}^{(b)}), \quad \mathbf{p}^{(b)} = \text{softmax}(\tilde{\mathbf{r}}^{(b)}), \quad (2.14)$$

where  $p_i^{(b)}$  is the initial routing weight for expert  $i$ . The adjusted scores  $\tilde{\mathbf{r}}^{(b)}$  may include expert bias, training noise or temperature scaling before the routing softmax. *Top-k routing* retains only the highest-scoring experts and zeros out all others. *Capacity constraints* prevent one expert from receiving too many assignments. If  $\mathcal{S}_b = \text{TopK}(\mathbf{p}^{(b)}, k)$  is the selected expert set, the sparse expert output can be written as

$$\mathbf{y}^{(b)} = \sum_{i \in \mathcal{S}_b} \alpha_i^{(b)} E_i(\mathbf{h}^{(b)}), \quad \alpha_i^{(b)} = \frac{p_i^{(b)}}{\sum_{j \in \mathcal{S}_b} p_j^{(b)}}. \quad (2.15)$$

Only the selected experts are evaluated, so the activated computation can remain small even when the total number of expert parameters is large. The same abstraction is useful for teacher routing: each teacher can be treated as an expert that contributes a weighted distillation signal for a subset of samples.

Sparse routing also needs mechanisms that prevent collapse onto a small subset of experts. In a sparse MoE layer, collapse is not only a modeling problem but also a systems problem: the overloaded experts determine the step latency, while under-used experts receive few gradients and remain poorly specialized [36, 39, 40]. Let a minibatch contain  $T$  routed items and let  $N$  denote the number of experts. For item  $b$ , the router produces probabilities  $\mathbf{p}^{(b)} \in \Delta^{N-1}$ . A top- $k$  router induces the binary assignment mask

$$\mathbf{a}_i^{(b)} = \mathbf{1}\{i \in \text{TopK}(\mathbf{p}^{(b)}, k)\}, \quad n_i = \sum_{b=1}^T \mathbf{a}_i^{(b)}, \quad (2.16)$$

where  $n_i$  is the number of routed items assigned to expert  $i$ . Balanced routing corresponds to  $n_i \approx kT/N$ . When  $n_i$  is much larger than this value for a few experts, the router has effectively reduced the model to a small ensemble of busy experts while wasting the capacity of the remaining ones. The routing statistics used by many auxiliary losses can be written as a *soft* load and a *hard* load:

$$\bar{p}_i = \frac{1}{T} \sum_{b=1}^T p_i^{(b)}, \quad \bar{q}_i = \frac{1}{kT} \sum_{b=1}^T \mathbf{a}_i^{(b)}. \quad (2.17)$$

Here,  $\bar{p}_i$  measures how much probability mass the router gives to expert  $i$  before hard selection, whereas  $\bar{q}_i$  measures how often the expert is actually selected after sparse routing. The Switch Transformer balancing loss uses their scaled dot product [37]:

$$\mathcal{L}_{bal} = \alpha_{bal} N \sum_{i=1}^N \bar{p}_i \bar{q}_i. \quad (2.18)$$

The factor  $N$  keeps the magnitude approximately independent of the number of experts because the uniform solution gives  $\sum_i (1/N)(1/N) = 1/N$ . The hard load  $\bar{q}_i$  is not differentiable through the top- $k$  decision, but  $\bar{p}_i$  is differentiable. The loss therefore gives the router a gradient signal that discourages assigning high average probability to already busy experts. In this thesis, the GRACE router uses entropy and  $z$ -loss terms in place of a hard capacity constraint, so load balancing is understood here as a stabilizer for sparse teacher selection, not a replacement for the task or distillation loss.

Sparse routing becomes especially useful when the highest-utility expert depends on the input [36, 37]. In a generic sample-adaptive router, the selected expert set is a function of a sample representation:

$$\mathcal{S}_b = \text{TopK}(\text{softmax}(G_\phi(\mathbf{h}^{(b)})), k). \quad (2.19)$$

The output or loss is then formed only from experts in  $\mathcal{S}_b$ . This differs from fixed ensembling, where every expert contributes to every example with the same global weight. Sample-adaptive routing is especially relevant when inputs have different structure. In multimodal QA, one sample may require OCR and document layout reasoning, another may require chart interpretation and another may depend mainly on natural image recognition. A router can learn to give different experts or teachers different responsibility across these cases. The mathematical distinction can be stated compactly. A global weighting scheme uses one vector  $\boldsymbol{\pi} \in \Delta^{N-1}$  for the full training set:

$$\mathcal{L}^{(b)} = \sum_{i=1}^N \pi_i \mathcal{L}_i^{(b)}. \quad (2.20)$$

A sample-adaptive scheme uses  $w_i^{(b)}$ :

$$\mathcal{L}^{(b)} = \sum_{i=1}^N w_i^{(b)} \mathcal{L}_i^{(b)}, \quad w_i^{(b)} = 0 \text{ for } i \notin \mathcal{S}_b. \quad (2.21)$$

The sparsity constraint limits the number of active experts and makes the teacher-selection decision interpretable. It also reduces the chance that low-utility or irrelevant teachers dominate the update for a sample.

The same routing abstraction can be used for distillation by treating frozen teachers as experts. The router does not combine hidden activations. Instead, it combines teacher-guided losses. If  $T_i$  is teacher  $i$ , then teacher  $T_i$  contributes a loss  $\mathcal{L}_{KD}^{(b,i)}$  for sample  $b$ . A routed distillation loss takes the form

$$\mathcal{L}_{route}^{(b)} = \sum_{i \in \mathcal{S}_b} \omega_i^{(b)} \mathcal{L}_{KD}^{(b,i)}. \quad (2.22)$$

This view connects MoE selection to multi-teacher KD: teacher routing is a conditional computation mechanism over supervision signals. Unlike standard MoE, the experts here are frozen, so only the student and router parameters receive gradient updates. Routing in a distillation setting has two additional complications. First, teacher quality is not a fixed property of the teacher alone. It depends on the input, the current student state and the task objective. Second, a high router score does not guarantee that the selected teacher produces a gradient aligned with supervised learning. This means routing is useful but not a complete solution to teacher selection by itself [21, 41, 42].

## 2.5 Multi-Teacher Distillation

A pool of frozen teachers can be aggregated by assigning each teacher a distillation weight. Let  $\mathcal{T} = \{T_1, \dots, T_N\}$  be such a pool. For sample  $b$ , teacher  $T_i$  produces a per-teacher distillation loss  $\mathcal{L}_{KD}^{(b,i)}$ . In principle, one could average these losses uniformly,

$$\mathcal{L}_{KD}^{(b)} = \frac{1}{N} \sum_{i=1}^N \mathcal{L}_{KD}^{(b,i)}, \quad (2.23)$$

but this assumes every teacher is equally useful for every sample. In practice, different teachers may provide conflicting guidance because their predictions, specialized strengths and KD gradients need not agree on the same training sample. Some teachers may help the supervised objective on a given sample, while others may push the student in a different direction [41]. The sparse-routing framework introduced in Section 2.4 applies directly here by treating teachers as experts whose usefulness varies across samples [21–23]. More

generally, an adaptive multi-teacher objective can be written as

$$\mathcal{L}_{KD}^{(b)} = \sum_{i=1}^N w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}, \quad w_i^{(b)} \geq 0, \quad \sum_{i=1}^N w_i^{(b)} = 1. \quad (2.24)$$

Uniform averaging is the special case  $w_i^{(b)} = 1/N$ . A routed method lets the weights depend on the sample, the student state and the current teacher-selection policy. This is useful when teachers specialize differently: one teacher may score higher on documents, another on charts and another on natural scene text. The main risk is that adaptive weighting can collapse to one teacher [21–23].

Sparse routing does not remove the possibility of gradient conflict. Even after it selects a teacher subset, the remaining teachers may still induce KD gradients that disagree with the supervised objective. To see why, consider the standard combined distillation loss  $\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{KD}$ . This objective minimizes a supervised term and a teacher-guided term together. Because both terms share the same student parameters, the combined update direction depends on whether the gradients  $\nabla_{\theta} \mathcal{L}_{CE}$  and  $\nabla_{\theta} \mathcal{L}_{KD}$  agree or conflict. When these two gradient directions are aligned, the combined update reinforces both objectives. When they oppose each other, the resulting update is a compromise that may degrade one or both objectives. This is an instance of *multi-objective optimization*, where a single parameter update must serve competing objectives simultaneously [41, 42]. In multi-teacher distillation, the problem is compounded because each teacher contributes its own KD gradient. A routing mechanism can select a small teacher subset, but selection alone cannot guarantee that the remaining KD gradients support the supervised task update.

## 2.6 Related Work

Previous work on heterogeneous VLM distillation falls into four groups.

**Response-based and cross-tokenizer KD.** Response-based KD transfers teacher output distributions to the student and has been applied to language-model and multimodal compression [12, 13, 30, 43, 44]. Its supervision is task-facing, but standard KL or JS losses assume a shared vocabulary. Tokenizer-agnostic and optimal-transport methods relax this by comparing probability mass over decoded strings [17–19, 31], yet are typically formulated for single-teacher transfer and do not address multi-teacher selection or weighting.

**Representation matching and CKA alignment.** Representation-level methods align hidden states or similarity structures, bypassing vocabulary mismatch entirely [20, 31]. However, hidden-state similarity is an indirect proxy for task behavior: high CKA does not guarantee matching answer probabilities after the language head, and these methods provide no mechanism for choosing among teachers.

**Multi-teacher KD and sparse routing.** Multi-teacher KD exploits teacher diversity for different input types [21–23], and sparse expert routing improves scalability by activating only a subset of teachers per sample [36, 37, 40]. The limitation is that routing estimates **relevance**, not **objective compatibility**: a routed teacher can still produce a KD gradient that conflicts with the supervised loss.

**Gradient-conflict resolution.** Gradient-based filtering directly measures whether auxiliary objectives agree with the supervised update direction [41, 42], but these methods are not designed as cross-tokenizer multi-teacher objectives and do not define a tokenizer-agnostic output loss.

Together, these groups leave a clear gap: no existing method jointly handles tokenizer mismatch, sample-adaptive teacher selection and objective-aware post-routing weighting. Addressing all three requires combining **tokenizer-agnostic output matching**, **sparse sample-dependent routing** and **gradient-agreement refinement** in a single framework.

# Chapter 3

## Proposed Method

This chapter presents Gradient-Routed Alignment with Capacity-aware Experts (GRACE) by first defining the heterogeneous multi-teacher distillation setting, then introducing the trie-Wasserstein KD loss, teacher routing network, gradient-agreement refinement mechanism and final training objective. The method targets vocabulary mismatch, variable teacher usefulness across samples and KD gradients that conflict with the supervised objective, with each component correcting one failure that the previous component cannot fix.

### 3.1 Method Overview and Problem Formulation

We consider multimodal understanding tasks that require reading image text, understanding the question and generating an answer sequence [4–6]. The training set is  $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N_{\text{data}}}$ , where  $x_n$  is a multimodal input and  $y_n$  is the target answer sequence. Let  $S_\theta$  denote the student VLM with trainable parameters  $\theta$ , let  $G_\phi$  denote the trainable teacher router with parameters  $\phi$  and let  $\mathcal{T} = \{T_i\}_{i=1}^N$  be the frozen teacher pool. For each sample  $b$ , the student produces logits  $\mathbf{z}_s^{(b)}$  and teacher  $T_i$  provides teacher logits  $\mathbf{z}_t^{(b,i)}$ .

This formulation treats teacher choice as part of the learning problem, not as a fixed preprocessing decision. A teacher specialized for document layout may provide useful supervision for forms and receipts, while a different teacher may be more suitable for natural scene text or chart reasoning. At the same time, teacher logits cannot be compared to student logits by token index when their tokenizers differ [17]. The method must therefore solve three coupled problems during training: compare teacher and student output distributions across different vocabularies, select a small set of relevant teachers for each sample and suppress teacher signals whose gradients conflict with the supervised answer objective [41, 42]. GRACE addresses these problems by combining tokenizer-agnostic trie-Wasserstein matching, student-conditioned sparse teacher routing and gradient-agreement refinement. Figure 3.1 summarizes the full training pipeline. The left block contains the student, the frozen teacher pool and the sample-level quantities produced by the forward pass: student logits, hidden states and the supervised cross-entropy loss. The middle block contains the two mechanisms that produce candidate teacher supervision. Trie-Wasserstein KD computes a per-teacher distillation loss over a shared byte trie, enabling tokenizer-agnostic output matching even when teacher and student vocabularies are different. In parallel, the teacher routing network maps pooled student hidden states to sparse routing weights and selects a small subset of candidate teachers for the current sample. The

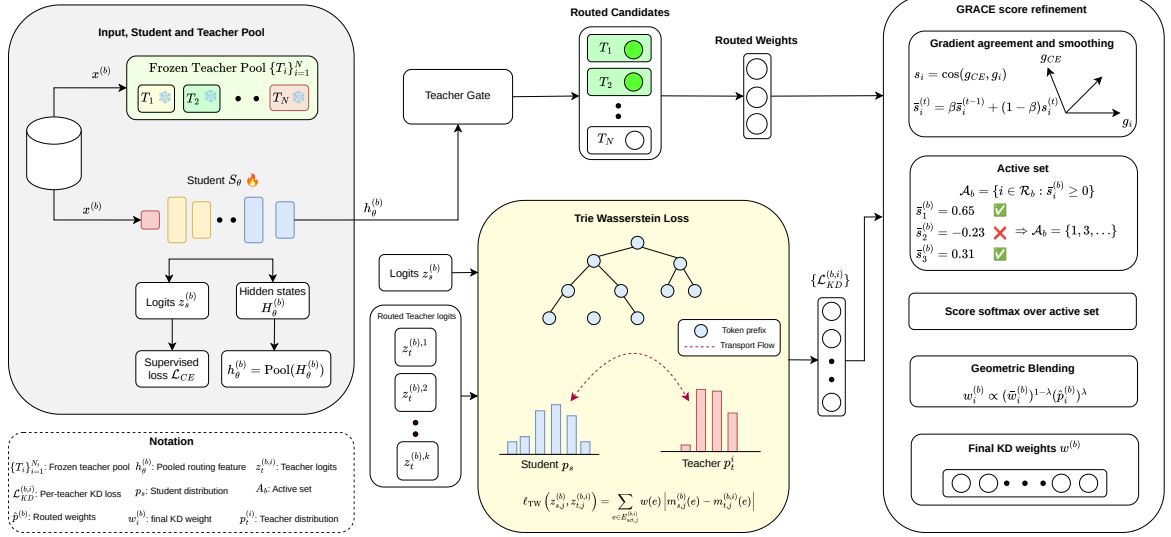


Figure 3.1: Overview of the GRACE architecture.

right block applies gradient-agreement refinement. It compares each routed teacher’s KD gradient with the supervised CE gradient, smooths the agreement signal and converts the routed scores into final KD weights.

The method is organized around four components. Trie-Wasserstein KD computes per-teacher distillation losses over a shared byte trie, so tokens are matched by decoded lexical structure, not by vocabulary index. The teacher routing network selects a sparse subset of candidate teachers from student-side hidden states at each training step. Gradient-agreement refinement adjusts the routed teacher weights based on how well each teacher’s distillation gradient aligns with the supervised objective. Finally, the training objective combines supervised cross-entropy, routed KD and router auxiliary losses into a single end-to-end update. Together, they extend sparse expert routing and token-level logit supervision to the heterogeneous multi-teacher VLM setting, where teachers and student may use different tokenizers and vocabularies [36, 37, 41].

A GRACE update starts from a mini-batch  $\mathcal{B} = \{(x_b, y_b)\}_{b=1}^B$  and a frozen teacher pool  $\mathcal{T} = \{T_i\}_{i=1}^N$ . The student first produces the logits, hidden states and supervised cross-entropy loss for the answer tokens. The router then uses the pooled student hidden state to select a sparse routed set  $\mathcal{R}_b$  for each sample. For every selected teacher, trie-Wasserstein KD computes a per-teacher loss  $\mathcal{L}_{KD}^{(b,i)}$ , using the shared byte-trie geometry to compare student and teacher output distributions across different tokenizers. Gradient-agreement refinement then filters and reweights the routed teachers by comparing each teacher-specific KD gradient with the supervised CE gradient. At a high level, the routed KD signal for

sample  $b$  is the weighted combination of teacher-specific losses:

$$\bar{\mathcal{L}}_{KD}^{(b)} = \sum_{i \in \mathcal{R}_b} w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}.$$

The final weights  $w_i^{(b)}$  are determined by both the router and the gradient-agreement refinement step. This decomposition separates teacher selection from teacher validation: routing proposes teachers that appear relevant to the sample, while gradient agreement checks whether their current KD directions are compatible with the supervised learning objective. The resulting KD term is then combined with CE and router regularization in one optimization objective. The main notation used throughout the method is listed in Table 3.1.

Table 3.1: Main notation.

| Symbol                                                                                                          | Meaning                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N_{\text{data}}}$                                                          | Multimodal samples and target answer sequences.                                                                   |
| $S_\theta, \mathcal{T} = \{T_i\}_{i=1}^N$                                                                       | Trainable student VLM and frozen teacher pool.                                                                    |
| $G_\phi$                                                                                                        | Trainable teacher routing network with parameters $\phi$ .                                                        |
| $T_{\text{train}}$                                                                                              | Total number of training steps.                                                                                   |
| $\mathbf{z}_s^{(b)}, \mathbf{z}_t^{(b,i)}$                                                                      | Student and teacher logits for sample $b$ .                                                                       |
| $\mathbf{H}_s, \mathbf{h}$                                                                                      | Student hidden states and pooled routing feature.                                                                 |
| $\rho_{\text{trie}}, \eta_{\text{boundary}}, K_{\text{TW}}, e_{\text{tail}}$                                    | Trie edge decay, boundary-edge weight, sparse support size and residual tail edge.                                |
| $k_{\text{route}}$                                                                                              | Configured router top- $k$ .                                                                                      |
| $\mathbf{r}, \mathbf{p}$                                                                                        | Router logits and soft router weights.                                                                            |
| $\mathcal{C}_b, \mathcal{R}_b, \hat{\mathbf{p}}^{(b)}$                                                          | Top- $k$ candidate teacher set, routed teacher set and renormalized routed weights.                               |
| $s_i, \hat{s}_i, \tilde{a}_i, a_i^{(b)}, \mathcal{A}_b$                                                         | Raw agreement score, EMA-smoothed agreement score, teacher active flag, sample active flag and active routed set. |
| $\omega^{(b)}, \mathbf{w}^{(b)}$                                                                                | Score-softmax weights and final KD weights.                                                                       |
| $\Omega_b^{CE}, \Omega_{b,i}^{KD}$                                                                              | Supervised CE positions and matched KD positions.                                                                 |
| $\mathbf{g}_{CE}, \mathbf{g}_i$                                                                                 | CE and teacher-KD parameter gradients used by GRACE.                                                              |
| $\beta_{\text{EMA}}, \beta_{\text{score}}, \epsilon_{\text{num}}, \epsilon_{\text{fb}}, \lambda_{\text{blend}}$ | EMA decay, score-softmax sharpness, cosine stabilizer, fallback threshold and router-score blend exponent.        |
| TopKRoute                                                                                                       | Sparse top- $k$ teacher routing with routed-weight renormalization.                                               |
| Uniform fallback                                                                                                | Uniform weights over the active routed set.                                                                       |
| Blend                                                                                                           | Mixing routed weights with GRACE score weights.                                                                   |
| EMA                                                                                                             | Exponential moving average for GRACE scores.                                                                      |

## 3.2 Trie-Wasserstein Cross-Tokenizer Distillation

GRACE applies distillation only on supervised answer tokens, not on prompt tokens. For sample  $b$ , let  $\Omega_b^{CE}$  denote the student positions whose labels are not ignored. For teacher  $T_i$ , let  $\Omega_{b,i}^{KD}$  denote the matched answer positions after masking ignored labels on both student and teacher sides and truncating both spans to a common length. This is a position-wise approximation across tokenizers: after answer-span preprocessing, the  $j$ -th student and teacher answer positions are treated as corresponding positions. Trie-Wasserstein matching handles the token-distribution mismatch at each aligned position, while sequence-level byte alignment remains outside the objective. Restricting distillation to answer positions avoids wasting capacity on prompt formatting, padding and instruction wording, where teacher and student distributions carry no task-relevant signal.

### 3.2.1 Shared Byte Trie

Direct token-index matching assumes that the  $k$ -th coordinate of the teacher distribution and the  $k$ -th coordinate of the student distribution refer to the same decoded token. This assumption fails when tokenizers differ: the same vocabulary index may correspond to unrelated token strings in the teacher and student vocabularies. GRACE maps decoded teacher and student token strings into a shared UTF-8 byte trie, so tokens live in a shared prefix tree, not arbitrary vocabulary-index space. Tokens with common byte prefixes share the same initial trie path even if their tokenizer IDs are unrelated, allowing probability mass from different vocabularies to be compared through shared decoded-string structure. For each sample  $b$ , token position  $j \in \Omega_{b,i}^{KD}$  and teacher  $T_i$ , let  $\mathbf{z}_{s,j}^{(b)} \in \mathbb{R}^{V_s}$  be the student logits and  $\mathbf{z}_{t,j}^{(b,i)} \in \mathbb{R}^{V_i}$  be the teacher logits. The student and teacher probability distributions are

$$\mathbf{p}_{s,j}^{(b)} = \text{softmax}\left(\frac{\mathbf{z}_{s,j}^{(b)}}{\tau_s}\right), \quad \mathbf{p}_{t,j}^{(b,i)} = \text{softmax}\left(\frac{\mathbf{z}_{t,j}^{(b,i)}}{\tau_t}\right), \quad (3.1)$$

where  $\tau_s$  and  $\tau_t$  are the student and teacher temperatures. In the sparse computation, ignored token IDs are masked before normalization so special and extended tokens do not receive probability mass in the trie.

Before insertion, each tokenizer piece is canonicalized into lexical content and optional boundary metadata. Visible boundary markers such as GPT-2-style space markers, SentencePiece space markers, WordPiece `##` markers and BPE `</w>` markers are stripped from the content string and recorded as boundary type. If the tokenizer decoder exposes leading whitespace directly, GRACE records that whitespace as a boundary signal and stores the stripped content bytes. The content is decoded, Unicode normalized and encoded as UTF-8 bytes. This keeps tokenizer conventions from splitting the shared

content path at the root. Marker-only pieces remain explicit content pieces, so canonicalization never removes a token from the trie.

Each canonical content string is terminated with an EOS sentinel. The sentinel separates complete tokens from strict prefixes, so `doc` and `document` remain distinct leaves even though they share an initial path. If a tokenizer boundary marker was removed during canonicalization, GRACE adds a boundary edge after the EOS sentinel with weight  $\eta_{\text{boundary}}$ . This design shifts the comparison away from identical token IDs and keeps probability mass on similar decoded strings close, while treating whitespace and word-boundary conventions as small tokenizer-specific differences.

The trie defines a geometry over decoded token strings, illustrated in Figure 3.2 for the pairs `document/receipt` and `doc/rece`. That geometry is well suited to OCR-heavy VQA, where correct and near-correct answers often differ through prefixes, suffixes, spaces or number formats. For example, `<space>cat`, `cat` and `##cat` share the same content bytes for `cat`, with boundary edges recording tokenizer conventions separately. For a normal content-byte or EOS edge  $e$ , let  $d(e)$  be its depth. GRACE assigns edge weights as follows:

$$w(e) = \begin{cases} \rho_{\text{trie}}^{d(e)-1}, & e \in E_{\text{content}} \cup E_{\text{EOS}}, \\ \eta_{\text{boundary}}, & e \in E_{\text{boundary}}. \end{cases} \quad (3.2)$$

where  $0 < \rho_{\text{trie}} < 1$  and  $\eta_{\text{boundary}} \geq 0$ . Here  $E_{\text{boundary}}$  contains optional marker edges for space, continuation and end-of-word metadata. The first content edge from the root has weight 1 and deeper content edges decay geometrically. This makes shared prefixes cheaper to transport across than unrelated byte paths, while boundary edges keep space and continuation markers close to their underlying content. The loss therefore penalizes how far the student’s probability mass must move through the shared lexical tree to resemble the teacher’s mass, not whether the student assigns probability to the same token indices. The weighting still preserves a valid tree metric [18, 33] and different tokenizers may stop at different depths while still sharing the same byte-prefix edges whenever their decoded token strings overlap in byte-prefix structure.

### 3.2.2 Sparse Support and Residual Tail Edge

The complete shared trie can compare every vocabulary token at every answer position, but this is expensive. GRACE builds a sparse augmented trie, so most useful distributional information, concentrated in high-probability tokens, is captured exactly. The tail edge handles the remainder: without it, omitted mass would silently disappear, making the teacher appear more confident than it really is and distorting the transport distance. At each aligned answer position, GRACE masks ignored vocabulary entries, retains the

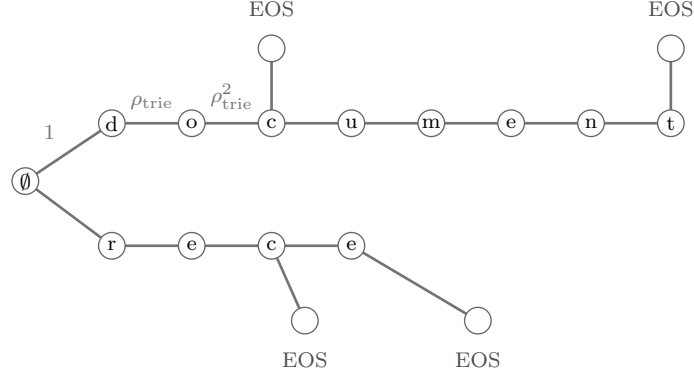


Figure 3.2: Shared-trie tokenizer matching.

top- $K_{\text{TW}}$  student and teacher tokens and inserts those tokens through their canonical content paths, followed by EOS and any optional boundary edge. Probability mass outside the retained sets is not discarded, but collapsed into a single residual tail edge. This gives an exact distance on the collapsed sparse trie and it approximates the complete trie distance by representing all omitted vocabulary mass with one tail atom.

Let  $K_{\text{TW}}$  denote the sparse trie support size. Let  $\mathcal{V}_s^*$  and  $\mathcal{V}_t^*$  denote the non-ignored student and teacher vocabularies. Before constructing the sparse trie, the student and teacher probabilities are renormalized over the non-ignored vocabularies so that the retained and residual masses form valid distributions. Operationally, ignored entries are excluded from the softmax normalizer before top- $K_{\text{TW}}$  selection. Let  $\mathcal{K}_{s,j}^{(b)} \subset \mathcal{V}_s^*$  and  $\mathcal{K}_{t,j}^{(b,i)} \subset \mathcal{V}_t^*$  be the retained token sets. The residual mass assigned to the tail edge is

$$r_{s,j}^{(b)} = 1 - \sum_{v \in \mathcal{K}_{s,j}^{(b)}} p_{s,j}^{(b)}(v), \quad r_{t,j}^{(b,i)} = 1 - \sum_{v \in \mathcal{K}_{t,j}^{(b,i)}} p_{t,j}^{(b,i)}(v). \quad (3.3)$$

Let  $\pi(v)$  be the trie path for token  $v$ , including canonical content edges, the EOS edge and any optional boundary edge. Let  $e_{\text{tail}}$  be the global residual tail edge with  $w(e_{\text{tail}}) = 1$ . For any active trie edge  $e$ , the sparse subtree masses are

$$m_{s,j}^{(b)}(e) = \sum_{\substack{v \in \mathcal{K}_{s,j}^{(b)} \\ e \in \pi(v)}} p_{s,j}^{(b)}(v) + \mathbf{1}[e = e_{\text{tail}}] r_{s,j}^{(b)}, \quad (3.4)$$

$$m_{t,j}^{(b,i)}(e) = \sum_{\substack{v \in \mathcal{K}_{t,j}^{(b,i)} \\ e \in \pi(v)}} p_{t,j}^{(b,i)}(v) + \mathbf{1}[e = e_{\text{tail}}] r_{t,j}^{(b,i)}. \quad (3.5)$$

### 3.2.3 Trie-Wasserstein Objective

The shared byte trie induces a tree metric over decoded token strings. On the complete trie, let  $\mathcal{V}_s$  and  $\mathcal{V}_t$  denote the student and teacher token vocabularies being compared

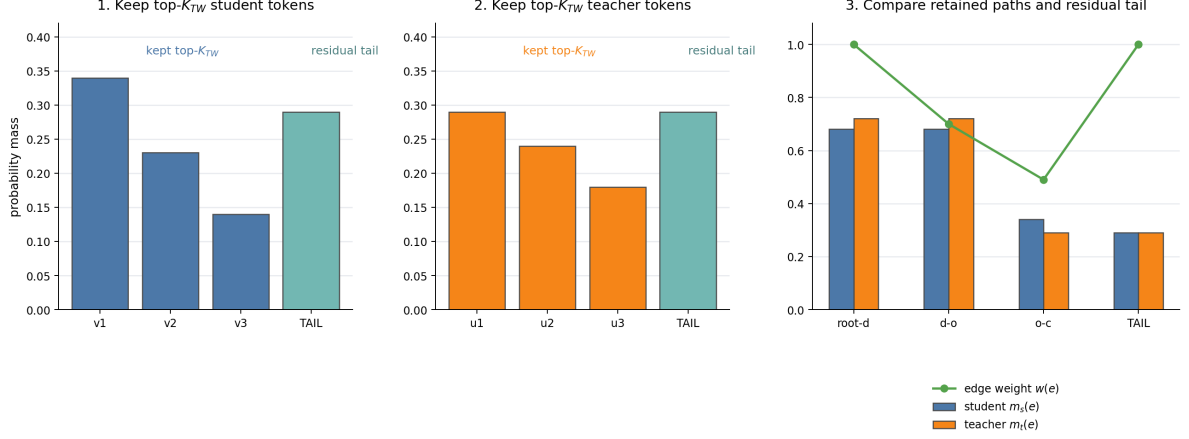


Figure 3.3: Sparse trie-Wasserstein support.

at a given answer position and let  $\pi(v)$  denote the trie path of decoded token  $v$ . For an edge  $e$ , the student and teacher subtree masses are

$$m_s(e) = \sum_{\substack{v \in \mathcal{V}_s \\ e \in \pi(v)}} p_s(v), \quad m_t(e) = \sum_{\substack{u \in \mathcal{V}_t \\ e \in \pi(u)}} p_t(u).$$

$m_s(e)$  and  $m_t(e)$  measure how much probability mass each distribution assigns to token paths passing through edge  $e$ . The  $W_1$  optimal transport distance between teacher and student token distributions on the trie can then be written as a weighted sum of subtree-mass imbalances:

$$W_T(p_s, p_t) = \sum_{e \in E} w(e) |m_s(e) - m_t(e)|. \quad (3.6)$$

This form follows from the standard  $W_1$  Kantorovich optimal transport problem under the trie-induced tree metric [18, 33]. The full proof is provided in Appendix A. In training, GRACE evaluates the same edgewise formula on the sparse active edge set defined above, using the retained token paths and residual tail edge. The token-level trie-Wasserstein loss is

$$\ell_{TW}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}) = \sum_{e \in \mathcal{E}_{act,j}^{(b,i)}} w(e) |m_{s,j}^{(b)}(e) - m_{t,j}^{(b,i)}(e)|, \quad (3.7)$$

where  $\mathcal{E}_{act,j}^{(b,i)}$  contains the active canonical content edges, EOS edges, optional boundary edges and the residual tail edge from the student and teacher supports.

The per-sample, per-teacher KD loss is the mean over supervised answer tokens:

$$\mathcal{L}_{KD}^{(b,i)} = \frac{1}{|\Omega_{b,i}^{KD}|} \sum_{j \in \Omega_{b,i}^{KD}} \ell_{TW}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}). \quad (3.8)$$

The complexity of sparse trie-Wasserstein computation has two parts: one-time trie construction and per-position sparse accumulation. The shared byte trie is built once

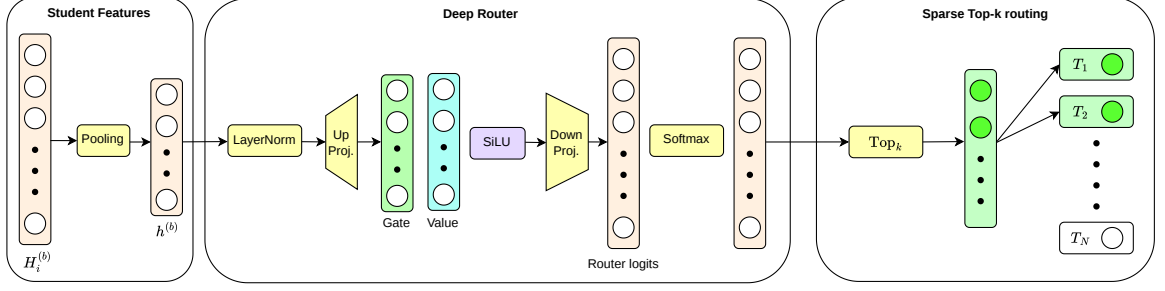


Figure 3.4: Teacher routing network.

with time and memory cost  $O(V_s \bar{L}_s + V_t \bar{L}_t)$ , where  $V_s$  and  $V_t$  are the student and teacher vocabulary sizes and  $\bar{L}_s$  and  $\bar{L}_t$  are the average canonical path lengths, including EOS and optional boundary edges. For one aligned answer position, extracting the top- $K_{\text{TW}}$  student and teacher tokens costs  $O(V_s \log K_{\text{TW}} + V_t \log K_{\text{TW}})$  with heap-based selection. Given the retained token sets, sparse subtree-mass accumulation walks only the retained canonical token paths and the residual tail edge, costing  $O(K_{\text{TW}} \bar{L}_s + K_{\text{TW}} \bar{L}_t + A)$ , where  $A$  is the number of active trie edges. If the implementation explicitly sorts or deduplicates active edge indices, this adds an  $O(A \log A)$  term. Otherwise, indexed accumulation gives an  $O(A)$  edge-aggregation cost. Runtime memory is  $O(E + P_s + P_t + A)$ , where  $E$  is the number of trie edges and  $P_s$  and  $P_t$  are the stored student and teacher path-index arrays. Unlike dense optimal transport, the sparse computation does not materialize a  $|V_s| \times |V_t|$  transport matrix.

### 3.3 Teacher Routing Network

With multiple teachers, uniform averaging ignores sample-specific teacher strengths: a document question, a chart question and a scene-text question may each benefit from a different expert and the student’s own hidden states are the most direct signal of which kind of example is being processed. This makes GRACE routing different from sparse mixture-of-experts routing in standard transformers: MoE routing chooses which expert computes the forward pass, while GRACE routing chooses which frozen external teacher shapes the student’s learning signal. GRACE therefore attaches a teacher routing network to the linear language-model head, so the gate operates on the text hidden state immediately before token prediction. The full pipeline from pooled student features to sparse top- $k$  teacher selection is shown in Figure 3.4 [21, 36, 37].

Let  $\mathbf{H}_s^{(b)}$  denote the student hidden states captured immediately before the language-model head for sample  $b$ . These are pooled over supervised answer positions to form the routing feature  $\mathbf{h}^{(b)} = \text{Pool}(\mathbf{H}_s^{(b)})$ . Pooling over answer positions captures what the student currently represents about the output, which is more informative than pooling over the full sequence. The routing feature is also student-state-dependent, not only dataset-dependent:

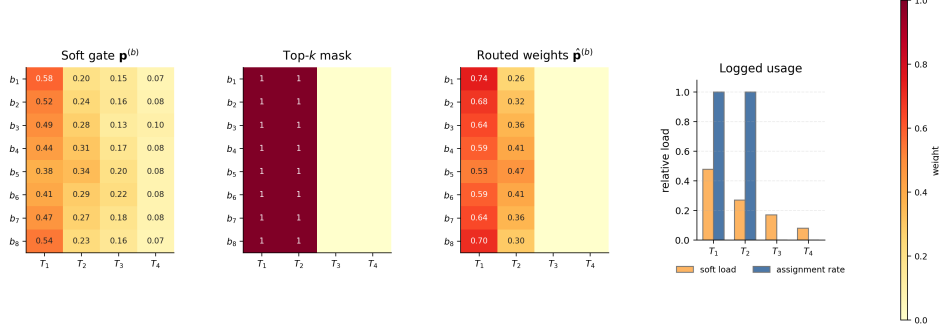


Figure 3.5: Top- $k$  routing dynamics.

the same image-question pair can produce a different routing decision as the student’s representations evolve. Pooling uses the mean over supervised answer positions.

A learned SwiGLU-style MLP router  $G_\phi(\cdot)$  maps the routing feature to gate logits  $\mathbf{r}^{(b)} = G_\phi(\mathbf{h}^{(b)})$ . These logits are converted to soft routing weights by  $\mathbf{p}^{(b)} = \text{softmax}(\mathbf{r}^{(b)})$ , where  $p_i^{(b)}$  is the initial gate weight for teacher  $T_i$ . The router then masks to the top- $k_{\text{route}}$  teachers and renormalizes the selected weights. The internal stack uses LayerNorm, an up-projection, a gate-value split, SiLU gating and a down-projection, keeping score computation dense while teacher use remains sparse after top- $k$  masking [36, 37].

The soft routing weights are dense, but distillation should not use every teacher for every sample. GRACE converts them into a sparse routed set by selecting the highest-scoring teachers for sample  $b$ ,  $\mathcal{C}_b = \text{TopK}(\mathbf{r}^{(b)}, \min(k_{\text{route}}, N))$ . The candidate set  $\mathcal{C}_b$  serves as the routed teacher set  $\mathcal{R}_b$ . With  $m_i^{(b)} = \mathbf{1}[i \in \mathcal{C}_b]$  denoting the top- $k$  assignment mask, non-selected weights are zeroed and the remaining softmax weights are renormalized:

$$\hat{p}_i^{(b)} = \frac{m_i^{(b)} p_i^{(b)}}{\sum_{j=1}^N m_j^{(b)} p_j^{(b)}}. \quad (3.9)$$

Top- $k$  routing makes supervision sparse and sample-specialized. Routing alone is not sufficient, however. A teacher can look relevant because its domain matches the input while its gradient still conflicts with the supervised target. For example, a document-specialized teacher may assign high probability to a plausible but incorrect field value. Gradient-agreement refinement, introduced in Section 3.4, addresses this residual failure. Figure 3.5 contrasts dense gate probabilities with the sparse routed assignment. The logged assignment rate records how often each teacher appears in the routed set.

Two auxiliary terms regularize teacher selection. The entropy term is implemented as a negative-entropy regularizer that discourages premature routing collapse [36, 37]:

$$\mathcal{L}_{\text{ent}} = -\frac{1}{B} \sum_{b=1}^B H(\mathbf{p}^{(b)}). \quad (3.10)$$

The router  $z$ -loss keeps router logits numerically well behaved:

$$\mathcal{L}_z = \frac{1}{B} \sum_{b=1}^B \left( \log \sum_{i=1}^N e^{r_i^{(b)}} \right)^2. \quad (3.11)$$

Both terms are pure stabilizers: entropy discourages the router from collapsing onto a single teacher too early, while  $z$ -loss keeps routing logits numerically controlled. Neither competes with the CE or KD objectives.

### 3.4 Gradient-Agreement Refinement

Routed teachers may still conflict with the supervised objective after teacher selection. Gradient-agreement refinement addresses this by scoring each teacher’s KD direction against the supervised CE direction and using those scores inside the routed candidate set. Routing identifies which teachers seem relevant to the current sample, while gradient agreement determines which of those candidates is currently compatible with the supervised learning objective. The two checks are complementary because domain relevance does not guarantee gradient compatibility, as discussed in Chapter 2.

To compute gradient agreement, let  $\Theta \subseteq \theta$  denote the trainable student parameters used by the optimizer. GRACE scores each teacher by comparing its teacher-specific KD gradient against the supervised CE gradient in student-parameter space. The CE gradient is  $\mathbf{g}_{CE} = \nabla_{\Theta} \mathcal{L}_{CE}$ . For teacher  $T_i$ , the KD gradient comes from that teacher’s mean distillation loss over the full mini-batch:  $\mathbf{g}_i = \nabla_{\Theta} (B^{-1} \sum_{b=1}^B \mathcal{L}_{KD}^{(b,i)})$ . Here,  $\mathcal{L}_{KD}^{(b,i)}$  is the trie-Wasserstein loss from Equation 3.8. These gradients score teacher compatibility while the final parameter update is still computed from the full training objective. As established in Chapter 2, a teacher whose KD gradient opposes the CE gradient pulls the student away from the correct answer even when the router selected it confidently.

GRACE quantifies how well supervised learning and KD are directionally aligned using cosine similarity:

$$s_i = \frac{\langle \mathbf{g}_{CE}, \mathbf{g}_i \rangle}{\max(\|\mathbf{g}_{CE}\|_2 \|\mathbf{g}_i\|_2, \epsilon_{\text{num}})}. \quad (3.12)$$

The agreement score  $s_i$  is teacher-level within the current mini-batch and is broadcast across samples. Sample dependence enters through the routed set  $\mathcal{R}_b$ , routed weights  $\hat{\mathbf{p}}^{(b)}$  and active set  $\mathcal{A}_b$ . To combine teacher-level agreement with sample-specific routing, GRACE first converts the agreement score into a binary teacher flag  $\tilde{a}_i = \mathbf{1}[s_i > \delta]$ , where  $\delta$  is the GRACE threshold. This flag is then intersected with the routed teacher set for each sample:

$$a_i^{(b)} = \mathbf{1}[i \in \mathcal{R}_b] \tilde{a}_i, \quad \mathcal{A}_b = \{i \in \mathcal{R}_b : a_i^{(b)} = 1\}. \quad (3.13)$$

The intersection with  $\mathcal{R}_b$  keeps the search inside the routed candidates: routing narrows the pool first and gradient agreement filters within it. The threshold  $\delta$  is applied to the raw score  $s_i$ , while the EMA-smoothed score  $\hat{s}_i$  controls score-softmax weighting and fallback spread. Agreement scores act as non-differentiable routing statistics, so the final optimization step does not backpropagate through the gradient-agreement computation itself or treat the scoring procedure as an additional trainable path.

Teacher weights can change abruptly when agreement scores are estimated from a single mini-batch. GRACE stabilizes this step with warmup, EMA smoothing and score-softmax weighting. Let  $\rho_g \in [0, 1]$  denote the warmup ratio relative to the total number of training steps. During the first  $\rho_g T_{\text{train}}$  steps, routed gate weights are left unchanged,  $w_i^{(b)} = \hat{p}_i^{(b)}$ . Early in training, the student’s representations and the router’s assignments are both still changing, so gradient-based filtering applied too soon risks rejecting useful teachers because the agreement scores are not yet reliable. Warmup prevents the agreement mechanism from overfitting to early gradient noise and missing genuine teacher usefulness. After warmup, GRACE smooths routed teacher scores. For teacher  $T_i$ , the previous running exponential moving average of past batch means is  $\bar{s}_i^{\text{prev}}$ , and its decay is  $\beta_{\text{EMA}}$ . The smoothed score is  $\hat{s}_i = \beta_{\text{EMA}} \bar{s}_i^{\text{prev}} + (1 - \beta_{\text{EMA}}) s_i$ . EMA smoothing turns gradient agreement from a momentary vote into a reputation score. For nonempty active sets, smoothed scores are converted into a score-softmax weight vector over  $\mathcal{A}_b$ :

$$\omega_i^{(b)} = \frac{\exp(\beta_{\text{score}} \hat{s}_i) \mathbf{1}[i \in \mathcal{A}_b]}{\sum_{j=1}^N \exp(\beta_{\text{score}} \hat{s}_j) \mathbf{1}[j \in \mathcal{A}_b]}, \quad (3.14)$$

$\beta_{\text{score}}$  controls how sharply the highest-scoring active teacher is preferred, setting up the fallback and blending cases below.

With score-softmax weights defined, GRACE assigns final KD weights through three mutually exclusive cases for each sample. These cases prevent the refinement step from overreacting when agreement evidence is missing or weak.

If no active teacher exists, meaning  $\mathcal{A}_b = \emptyset$ , the routed weights are retained as  $w_i^{(b)} = \hat{p}_i^{(b)}$ . If active teachers exist but their smoothed scores are too close, uniform weights are used over the active set. The score range is  $\Delta^{(b)} = \max_{i \in \mathcal{A}_b} \hat{s}_i - \min_{i \in \mathcal{A}_b} \hat{s}_i$ . If  $\mathcal{A}_b \neq \emptyset$  and  $\Delta^{(b)} < \epsilon_{\text{fb}}$ , then  $w_i^{(b)} = \mathbf{1}[i \in \mathcal{A}_b] / |\mathcal{A}_b|$ .

When the score range is confident enough, router weights are blended with score weights. Let  $\lambda_{\text{blend}} \in [0, 1]$  be the router-blend exponent. The unnormalized blended weights are  $\tilde{w}_i^{(b)} \propto (\hat{p}_i^{(b)})^{\lambda_{\text{blend}}} (\omega_i^{(b)})^{1-\lambda_{\text{blend}}}$ , followed by normalization over the active routed teachers:

$$w_i^{(b)} = \frac{\tilde{w}_i^{(b)} \mathbf{1}[i \in \mathcal{A}_b]}{\sum_{j=1}^N \tilde{w}_j^{(b)} \mathbf{1}[j \in \mathcal{A}_b]}. \quad (3.15)$$

Equation 3.15 completes the gradient-refinement operator: the router chooses  $\mathcal{R}_b$ , agreement



Figure 3.6: Representative KD weighting regimes.

filtering narrows it to  $\mathcal{A}_b$  and final weights are retained from the router, assigned uniformly or geometrically blended depending on confidence. These regimes are shown in Figure 3.6. This design keeps teacher selection sparse throughout: gradient-agreement refinement can narrow the routed set but never expands it, so KD cost remains bounded by  $k_{\text{route}}$  teachers per sample.

### 3.5 Routed KD Aggregation and Final Objective

At the sample level, only teachers in the routed set  $\mathcal{R}_b$  contribute to the KD loss:

$$\bar{\mathcal{L}}_{KD}^{(b)} = \sum_{i \in \mathcal{R}_b} w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}, \quad (3.16)$$

where  $w_i^{(b)}$  reflects whichever of the three post-routing rules applies (fallback, uniform-active or blended). The mini-batch KD term averages these:  $\mathcal{L}_{KD} = B^{-1} \sum_{b=1}^B \bar{\mathcal{L}}_{KD}^{(b)}$ . Because each per-teacher loss uses trie-Wasserstein matching, this term stays valid across heterogeneous tokenizers without direct token-index KL divergence.

The complete GRACE objective is

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{KD} \mathcal{L}_{KD} + \lambda_{ent} \mathcal{L}_{ent} + \lambda_z \mathcal{L}_z, \quad (3.17)$$

where  $\mathcal{L}_{CE}$  anchors the student to supervised answer tokens,  $\mathcal{L}_{KD}$  transfers routed teacher supervision and  $\mathcal{L}_{ent}$  together with  $\mathcal{L}_z$  regularize the router. Only  $\theta$  and  $\phi$  are updated; all teacher parameters remain frozen.

# Chapter 4

## Evaluation Methodology

This chapter describes the experimental setup used to evaluate GRACE and all compared methods. It covers the student models, training configuration, benchmark datasets and their evaluation metrics, baselines and ablation settings needed to reproduce the main results.

### 4.1 Experimental Setup

All experiments follow a shared evaluation protocol for fair comparison across methods. The experiments instantiate the student side with **SmolVLM-500M** and **SmolVLM-256M**, evaluated on multimodal benchmarks [16]. All models are evaluated with **VLM-Eval-Kit** [45], which provides a shared inference, post-processing and metric-computation pipeline for the three benchmarks. VLM-Eval-Kit uses a generation-based evaluation interface. For each benchmark sample, the toolkit loads the image-question pair, applies the dataset-specific prompt construction logic and calls the model’s generation or chat interface to produce a free-form text answer.

Before training, all benchmark samples are converted into a unified **LLaVA-style multimodal instruction format** [3] as shown in listing 4.1, so TextVQA, DocVQA and ChartQA can share the same data loader, prompt template and answer supervision interface.

```
1 {  
2   "image": "sample_000123.jpg",  
3   "conversations": [  
4     {"from": "human", "value": "<image>\nWhat is the invoice number?"},  
5     {"from": "gpt", "value": "4572"}  
6   ]  
7 }
```

Listing 4.1: Representative LLaVA-style sample used after dataset conversion.

We run all training jobs on a single high-memory **NVIDIA H200** graphics processing unit (GPU) or on comparable single-GPU **Vast.ai** instances, depending on availability. Distillation training uses **DeepSpeed ZeRO Stage 2**, **Brain Floating Point 16 (bf16)** mixed precision, **TensorFloat-32 (TF32)** matrix multiplication and gradient checkpointing to control memory use while keeping the student fully trainable and all teachers frozen [46, 47]. Full optimization and routing parameter values are reported in Appendix B, Table B.1. Teacher-logit storage and remote caching are documented in



Figure 4.1: Representative dataset examples.<sup>1</sup>

Appendix B.2 and Appendix B.3 for reproducibility purposes.

## 4.2 Datasets

We evaluate three text-centric VQA benchmarks: **TextVQA**, **DocVQA** and **ChartQA**. These benchmarks test whether the model can read visual text, understand questions and perform basic reasoning across natural images, documents and charts. They were chosen because they cover complementary visual domains and combine OCR-heavy understanding with layout interpretation and numerical reasoning [4–6].

**TextVQA** tests question answering on natural images containing scene text, requiring the model to read and reason over text embedded in real-world scenes [4]. *Sample question*: “What word is written on the stop sign?”

**DocVQA** tests question answering on scanned document images, where layout, paragraphs, tables and forms are central to answering correctly [5]. *Sample question*: “What is the invoice number shown in this document?”

**ChartQA** tests question answering over charts, combining chart reading with numerical and logical reasoning [6]. *Sample question*: “Which year has the highest sales value in the chart?”

Figure 4.1 shows the visual differences behind this benchmark choice. The TextVQA panel represents natural-image reading, where the relevant cue is embedded in an object in the scene. The DocVQA panel is a receipt with fields, item rows and totals arranged by document layout. The ChartQA panel is a bar chart, where answering requires comparing bar heights and reading values from the plot structure. The evaluation therefore covers natural scene text, document layouts and chart reasoning. The split sizes used throughout the evaluation chapter are listed in Table 4.1.

<sup>1</sup>Images adapted from Wikimedia Commons sources [48–50].

Table 4.1: Dataset split sizes.

| Dataset                 | Train  | Val   | Test  | Total  |
|-------------------------|--------|-------|-------|--------|
| TextVQA                 | 34,602 | 5,000 | 5,734 | 45,336 |
| DocVQA                  | 39,463 | 5,349 | 5,188 | 50,000 |
| ChartQA-H (human)       | 7,398  | 960   | 1,250 | 9,608  |
| ChartQA-M (machine)     | 20,901 | 960   | 1,250 | 23,111 |
| ChartQA (H+M, combined) | 28,299 | 1,920 | 2,500 | 32,719 |

The official metrics are dataset-specific. For TextVQA, we report VQA Accuracy: for a predicted answer  $\hat{a}$ , the per-question score is  $s_{\text{VQA}} = \min(\#\text{annotators that provided } \hat{a}/3, 1)$ , and the final score is the mean over all questions [4]. For DocVQA, we report ANLS, the Average Normalized Levenshtein Similarity [5]:

$$\text{NLS}(a, \hat{a}) = 1 - \frac{d_{\text{Lev}}(a, \hat{a})}{\max(|a|, |\hat{a}|)}, \quad s_{\text{ANLS}} = \begin{cases} \text{NLS}(a, \hat{a}), & \text{NLS} \geq 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

For ChartQA, we report Relaxed Accuracy [6]. Text predictions must match exactly, while numeric predictions are scored with

$$s_{\text{ChartQA}} = \begin{cases} \mathbf{1}[\hat{y} = y], & y = 0, \\ \mathbf{1}\left[\frac{|\hat{y}-y|}{|y|} \leq 0.05\right], & y \neq 0. \end{cases}$$

### 4.3 Baselines

With the datasets and official metrics defined above, Table 4.2 organizes the compared methods into four groups: non-distillation references, same-family KD, tokenizer-mismatched single-teacher transfer and heterogeneous teacher methods.

The non-distillation references consist of the original student and the full-finetuning baseline, which provide student-only reference points. The same-family baselines use **SmolVLM2-2.2B-Instruct** as teacher for forward KL, reverse KL and Jensen-Shannon distillation, isolating standard teacher-student transfer when the tokenizer and output space are matched [12, 16, 29, 51, 52]. The different-family single-teacher setting uses ULD with a teacher outside the SmolVLM family, testing tokenizer-mismatched transfer without teacher routing [17]. The heterogeneous teacher setting compares uniform averaging, SHM-CKA, REINFORCE and GRACE over a mixed teacher pool, separating static averaging, hidden-state matching and adaptive teacher assignment [18, 22, 23, 31].

All trainable competitor methods retain the supervised term  $\mathcal{L}_{CE}$ , so their differences come from the added transfer signal: the logits-only baselines change only the KD term,

Table 4.2: Baseline groups.

| Group                           | Methods                                                    | Teacher setup                                | Purpose                                                                    |
|---------------------------------|------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------|
| Non-distillation references     | Original, full finetuning                                  | No teacher                                   | Student-only reference points                                              |
| Same-family baselines           | Forward KL, Reverse KL, Jensen-Shannon                     | SmolVLM2-2.2B-Instruct                       | Standard single-teacher objectives within the SmolVLM family               |
| Cross-family single-teacher     | ULD                                                        | Qwen2-VL-2B-Instruct                         | Cross-family transfer without routed teacher selection                     |
| Heterogeneous teacher baselines | Uniform averaging, SHM-CKA, REINFORCE, <b>GRACE (Ours)</b> | Mixed teacher set from multiple VLM families | Static averaging vs. hidden-state matching vs. adaptive teacher assignment |

SHM-CKA adds hidden-state alignment and REINFORCE adds a policy-gradient term [20, 21, 53]. In the descriptions below,  $\gamma$  denotes a generic KD mixing coefficient.  $\alpha_{\text{KD}}$  is reserved for the GRACE training objective as defined in Equation 3.17.

**Original.** Directly evaluates the released student without additional optimization, thereby measuring the native capability of the student model before task adaptation or any teacher-guided distillation signal.

**Full finetuning.** Optimizes only  $\mathcal{L}_{CE}$  and uses no teacher, which isolates the effect of supervised task adaptation from any teacher-guided distillation signal.

**Forward KL.** Uses  $\mathcal{L}_{KD} = \text{KL}(p_t \| p_s)$  for same-family soft-target distillation when teacher and student vocabularies are aligned [12].

**Reverse KL.** Uses  $\mathcal{L}_{KD} = \text{KL}(p_s \| p_t)$  [52, 54] as a student-centered divergence baseline, which tests whether matching the teacher from the student’s current distribution is suitable for VLM distillation. Because the reverse-KL gradient is weighted by  $p_s$ , this objective tends to emphasize regions where the student already assigns probability.

**Jensen-Shannon.** Uses the symmetric divergence

$$\frac{1}{2}\text{KL}(p_s \| m) + \frac{1}{2}\text{KL}(p_t \| m), \quad m = \frac{1}{2}(p_s + p_t).$$

It tests whether a bounded symmetric objective is more stable than KL [29, 52].

**ULD.** Uses  $\mathcal{L}_{KD} = \|\pi(p_s) - \pi(p_t)\|_1$  [17]. This is the main tokenizer-mismatched logits baseline because it compares ranked probability mass, not token IDs.

**SHM-CKA.** Extends ULD with a hidden-state CKA penalty [17, 20, 31]. For the heterogeneous setting, both the ULD and layer-matching terms are averaged uniformly across teachers before combining with  $\mathcal{L}_{CE}$ . Full implementation details are given in Appendix B.1.

**Uniform multi-teacher.** Assigns fixed equal weights to all teachers in the heterogeneous pool,  $\mathcal{L}_{KD} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{KD}^{(i)}$  [21, 22], thereby testing whether teacher diversity alone is sufficient without adaptive routing or gradient-aware weighting.

**REINFORCE.** Replaces deterministic routing with a stochastic policy that samples teachers and updates the selector with REINFORCE [21, 53], providing an adaptive routing baseline that does not use the gradient-agreement refinement introduced by GRACE. The selector is trained with the policy-gradient term

$$\mathcal{L}_{\text{pol}} = -(r - b) \log p(\mathbf{m} \mid \boldsymbol{\pi}), \quad (4.1)$$

where  $r$  is the reward,  $b$  is an EMA reward baseline and  $\mathbf{m}$  is the sampled teacher mask. Its final training objective is

$$\mathcal{L}_{\text{RS}} = \mathcal{L}_{CE} + \gamma \mathcal{L}_{KD} + \lambda_{\pi} \mathcal{L}_{\text{pol}}. \quad (4.2)$$

Additional baseline implementation details are reported in Appendix B.1.

## 4.4 Ablation Settings

The ablation study in Section 5.2 isolates the main components of GRACE on DocVQA while keeping the teacher set and training protocol close to the final method. The variants change one mechanism at a time so that the result table can separate teacher diversity, sparse routing, gradient-agreement weighting and EMA score smoothing.

**Cross-family ULD.** The single-teacher cross-family baseline uses one teacher outside the SmolVLM family and optimizes

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{\text{KD}} \mathcal{L}_{ULD}, \quad \mathcal{L}_{ULD} = \|\pi(p_s) - \pi(p_t)\|_1.$$

**Uniform averaging.** The uniform variant adds the full heterogeneous teacher pool but assigns fixed equal weights:

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{\text{KD}} \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{TW}^{(i)}.$$

This setting tests whether teacher diversity alone is enough without adaptive selection.

**Routing only.** The routing-only variant keeps sparse teacher selection but replaces post-routing weights with uniform weights within the routed set:

$$w_i^{(b)} = u_i^{(b)} = \frac{\mathbf{1}[i \in \mathcal{R}_b]}{|\mathcal{R}_b|}.$$

This setting tests whether top- $k$  routing is useful without gradient-agreement refinement.

**Full GRACE.** The full method combines trie-Wasserstein KD, sparse routing and gradient-agreement weighting:

$$w_i^{(b)} \propto (\hat{p}_i^{(b)})^{\lambda_{\text{blend}}} (\omega_i^{(b)})^{1-\lambda_{\text{blend}}}.$$

Two additional ablation variants isolate the final weighting mechanism, reported in Table 5.6. The  $\lambda_{\text{blend}} = 1.0$  variant keeps the active-set threshold and fallback rules but removes the score-softmax factor from the confident-score case:

$$w_i^{(b)} \propto \hat{p}_i^{(b)} \mathbf{1}[i \in \mathcal{A}_b].$$

This row is an active-set-restricted router-weight ablation, not a pure router-only ablation over all routed teachers. The  $\beta_{\text{EMA}} = 0$  variant removes temporal smoothing by setting  $\hat{s}_i = s_i$  before score-softmax weighting.

# Chapter 5

## Experiment Results

This chapter reports the empirical results for GRACE. Section 5.1 answers RQ1 by comparing GRACE with finetuning, same-family KD, cross-family ULD and heterogeneous multi-teacher baselines. Section 5.2 answers RQ2 through component ablations. Section 5.3 answers RQ3 by analyzing training, storage and computational overhead. The final section discusses the main observed trends.

### 5.1 Main Comparison Results

This section reports the main empirical comparison used to answer RQ1. The results are organized from simpler reference points to the full heterogeneous multi-teacher setting: original students, supervised fine-tuning, same-family logits KD, cross-family ULD and multi-teacher distillation methods. Before applying adaptation or distillation, we report the original performance of the two student models, **SmolVLM-500M** and **SmolVLM-256M**, using **VLM-Eval-Kit** [45] under the evaluation pipeline described in Chapter 4. The student-only starting point for all later fine-tuning and distillation comparisons is given in Table 5.1. The teacher-side reference range is reported in Table 5.2. The gap between these student models and the higher-scoring teachers motivates the distillation experiments that follow.

Table 5.1: Original SmolVLM student results.

| Model        | TextVQA | DocVQA | ChartQA |
|--------------|---------|--------|---------|
| SmolVLM-500M | 56.75   | 59.52  | 62.80   |
| SmolVLM-256M | 47.17   | 48.64  | 45.04   |

Table 5.2: Teacher and reference-model results.

| Teacher Model          | TextVQA | DocVQA | ChartQA |
|------------------------|---------|--------|---------|
| Qwen2.5-VL-3B-Instruct | 79.30   | 93.90  | 94.24   |
| InternVL3-1B           | 74.10   | 81.90  | 75.30   |
| Gemma-3-4B-it          | 57.80   | 75.80  | 63.60   |
| Qwen2-VL-2B-Instruct   | 79.70   | 90.10  | 89.28   |

Using the highest teacher score on each benchmark as the reference, the original teacher-

student gap is 22.95, 34.38 and 31.44 points for SmolVLM-500M on TextVQA, DocVQA and ChartQA. For SmolVLM-256M, the corresponding gaps are 32.53, 45.26 and 49.20 points. These gaps define the headroom for the gap-to-teacher analysis and motivate the full benchmark comparison. The main benchmark results across original models, direct fine-tuning, standard logits distillation baselines and GRACE are collected in Table 5.3. This main result table reports the highest-scoring completed run for each method and benchmark. Uniform, SHM-CKA, REINFORCE and GRACE use a four-teacher mixed set spanning the Qwen, InternVL and Gemma families. Chapter 4 documents the four-teacher configuration used across all multi-teacher experiments. We report the baselines in four groups. The non-distillation references are the original student and full-finetuning runs. The same-family baselines use **SmolVLM2-2.2B-Instruct** as teacher for forward KL, reverse KL and Jensen-Shannon divergence [12, 13, 31]. The different-family single-teacher setting uses ULD with a teacher outside the SmolVLM family [17]. The multi-teacher baselines compare four-teacher Uniform averaging with adaptive methods such as SHM-CKA, REINFORCE and GRACE [21–23, 53].

Table 5.3: Main comparison across TextVQA, DocVQA and ChartQA.

| Category                                                                                   | Method         | Teacher              | Student      | TextVQA | DocVQA | ChartQA |
|--------------------------------------------------------------------------------------------|----------------|----------------------|--------------|---------|--------|---------|
| Non-distillation references                                                                |                |                      |              |         |        |         |
| Original                                                                                   | None           | None                 | SmolVLM-500M | 56.75   | 59.52  | 62.80   |
|                                                                                            |                |                      | SmolVLM-256M | 47.17   | 48.64  | 45.04   |
| Reference                                                                                  | Finetuning     | None                 | SmolVLM-500M | 56.96   | 61.96  | 63.00   |
|                                                                                            |                |                      | SmolVLM-256M | 49.83   | 42.03  | 49.68   |
| Same-family distillation baselines                                                         |                |                      |              |         |        |         |
| Same-family                                                                                | Forward KL     | SmolVLM2-2.2B        | SmolVLM-500M | 59.00   | 63.55  | 64.52   |
|                                                                                            |                |                      | SmolVLM-256M | 50.88   | 49.46  | 56.80   |
|                                                                                            | Reverse KL     |                      | SmolVLM-500M | 44.73   | 29.28  | 30.16   |
|                                                                                            |                |                      | SmolVLM-256M | 29.75   | 23.28  | 17.20   |
|                                                                                            | Jensen-Shannon |                      | SmolVLM-500M | 42.33   | 36.43  | 43.75   |
|                                                                                            |                |                      | SmolVLM-256M | 33.84   | 33.12  | 50.93   |
| Different-family single-teacher baselines                                                  |                |                      |              |         |        |         |
| Cross-family                                                                               | ULD            | Qwen2-VL-2B-Instruct | SmolVLM-500M | 61.49   | 65.28  | 66.92   |
|                                                                                            |                |                      | SmolVLM-256M | 53.95   | 52.24  | 60.96   |
| Multi-teacher baselines                                                                    |                |                      |              |         |        |         |
| Multi-teacher                                                                              | Uniform        | †                    | SmolVLM-500M | 63.88   | 69.59  | 75.60   |
|                                                                                            |                |                      | SmolVLM-256M | 54.57   | 55.70  | 63.68   |
|                                                                                            | SHM-CKA        |                      | SmolVLM-500M | 60.47   | 63.32  | 76.88   |
|                                                                                            |                |                      | SmolVLM-256M | 55.93   | 57.24  | 62.40   |
|                                                                                            | REINFORCE      |                      | SmolVLM-500M | 63.26   | 64.00  | 70.27   |
|                                                                                            |                |                      | SmolVLM-256M | 56.31   | 58.53  | 68.74   |
|                                                                                            | GRACE (Ours)   |                      | SmolVLM-500M | 69.38   | 73.11  | 80.20   |
|                                                                                            |                |                      | SmolVLM-256M | 60.73   | 60.48  | 72.32   |
| † Teacher pool: Qwen2-VL-2B-Instruct, Qwen2.5-VL-3B-Instruct, InternVL3-1B, Gemma-3-4B-it. |                |                      |              |         |        |         |

† Teacher pool: Qwen2-VL-2B-Instruct, Qwen2.5-VL-3B-Instruct, InternVL3-1B, Gemma-3-4B-it.

On DocVQA, GRACE rises from 69.59 to 73.11 on SmolVLM-500M and from 55.70 to 60.48 on SmolVLM-256M relative to the Uniform baseline. ChartQA shows the largest margins over Uniform: 75.60 to 80.20 for SmolVLM-500M and 63.68 to 72.32 for SmolVLM-256M. Three consistent patterns emerge across all benchmarks. Forward KL gives small gains over the original students, while reverse KL reduces performance and Jensen-Shannon varies by task. Cross-family ULD scores above the same-family logits losses on all three tasks, indicating that tokenizer-mismatched distillation is more effective in this setting than vocabulary-matched KL. Uniform averaging adds another increase over ULD, with gains of +2.39, +4.31 and +8.68 for SmolVLM-500M on TextVQA, DocVQA and ChartQA. The

additional improvement from single-teacher ULD to GRACE comes from trie-Wasserstein KD, sample-dependent sparse routing and gradient-agreement refinement after routing. GRACE also improves on the uniform multi-teacher baseline for both student sizes. The gain is smaller on TextVQA and DocVQA than on ChartQA, indicating that adaptive weighting has the largest observed benefit when the task mixes visual reading, numerical reasoning and teacher specialization across datasets.

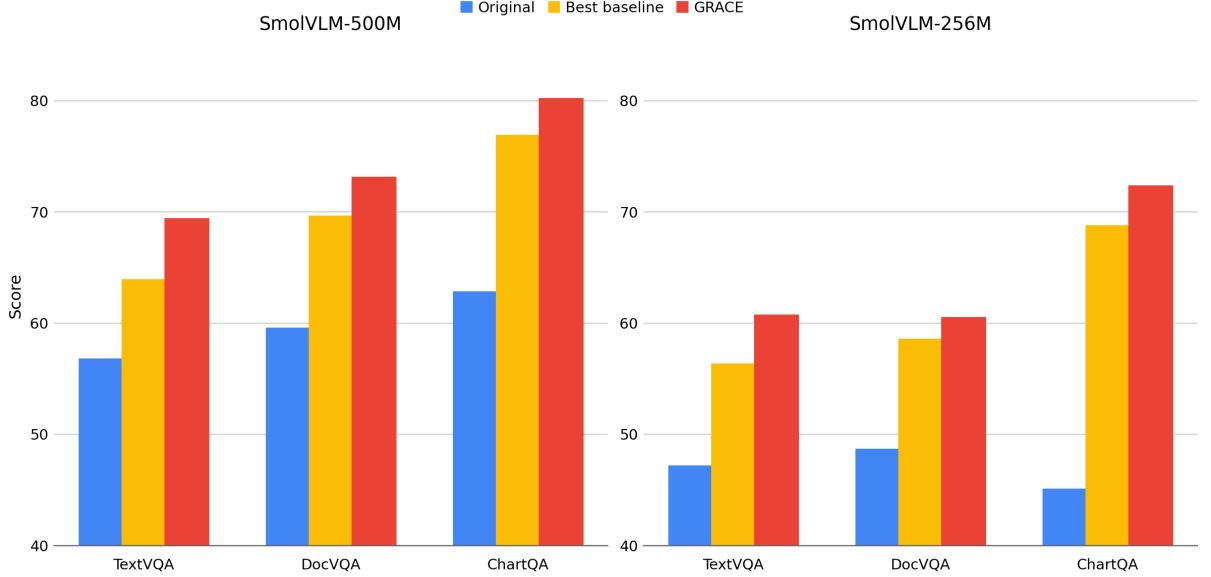


Figure 5.1: Grouped benchmark comparison.

To provide qualitative evidence of attention improvement, we use DEX-AR attribution maps [55]. For token  $t$ , layer  $l$  and head  $h$ , DEX-AR forms a gradient-based attribution map

$$\mathbf{G}_t^{(l,h)} = \nabla_{\mathbf{A}^{(l,h)}} y_t \odot \mathbf{A}^{(l,h)},$$

DEX-AR then keeps visually grounded heads with weights defined as

$$w^{(l,h)} = \max(S_{\text{img}}^{(l,h)} - S_{\text{text}}^{(l,h)}, 0).$$

The token-level and sentence-level attribution maps are defined as

$$\mathbf{H}_t = \sum_{l,h} w^{(l,h)} \mathbf{G}_t^{(l,h)}, \quad \mathbf{H}_{\text{sent}} = \sum_t \delta_t \mathbf{H}_t.$$

Here  $\delta_t = 1/|\Omega_{\text{ans}}|$  for answer tokens  $t \in \Omega_{\text{ans}}$ , where  $\Omega_{\text{ans}}$  is the set of generated answer token positions and  $\delta_t = 0$  otherwise. The sentence-level map is therefore the uniform average of token-level maps over the generated answer span. Raw attention maps and generic saliency methods are not well suited here because the model is autoregressive and the quantity of interest is which image regions drove the answer, not which regions are visually salient in general. DEX-AR is designed for this target: it attributes each answer

token back to the input image and averages those attributions across the answer span. Figure 5.2 summarizes the pipeline before the qualitative TextVQA example, making the later heatmaps easier to interpret as answer-grounding evidence, not generic saliency maps.

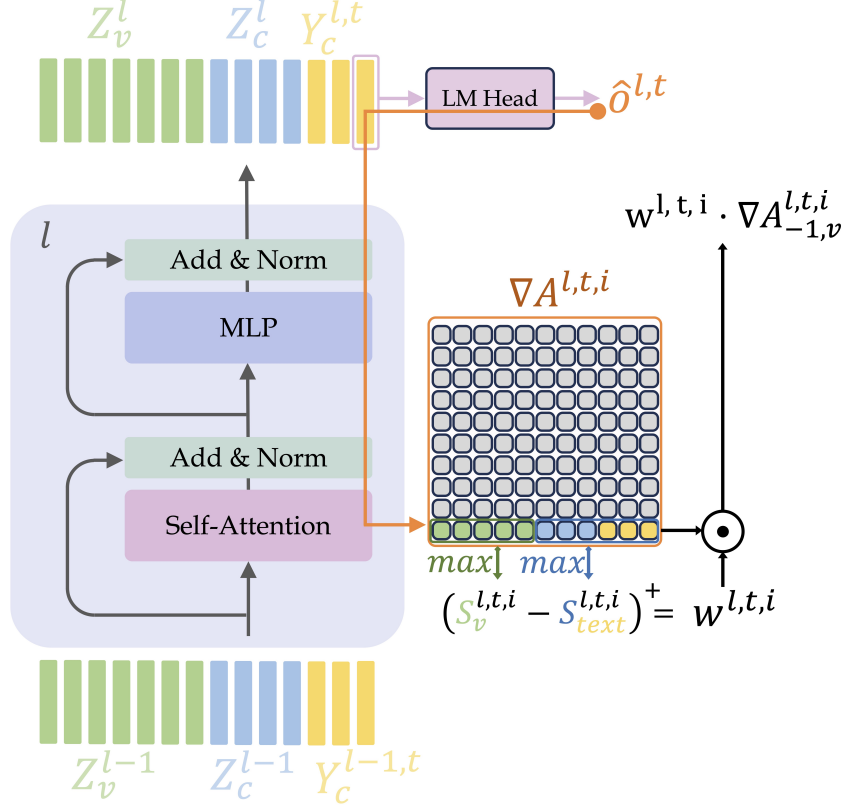


Figure 5.2: DEX-AR overview.

Figure 5.3 gives a representative TextVQA example. Sample 3 asks “*who is this copyrighted by?*” The relevant visual evidence is the copyright text on the lower-left border. The original SmolVLM-500M concentrates attention on the *Virgin* logo in Figure 5.3b, not this region. In contrast, GRACE redirects attention to the copyright text in Figure 5.3c, which is the relevant evidence for the correct answer.

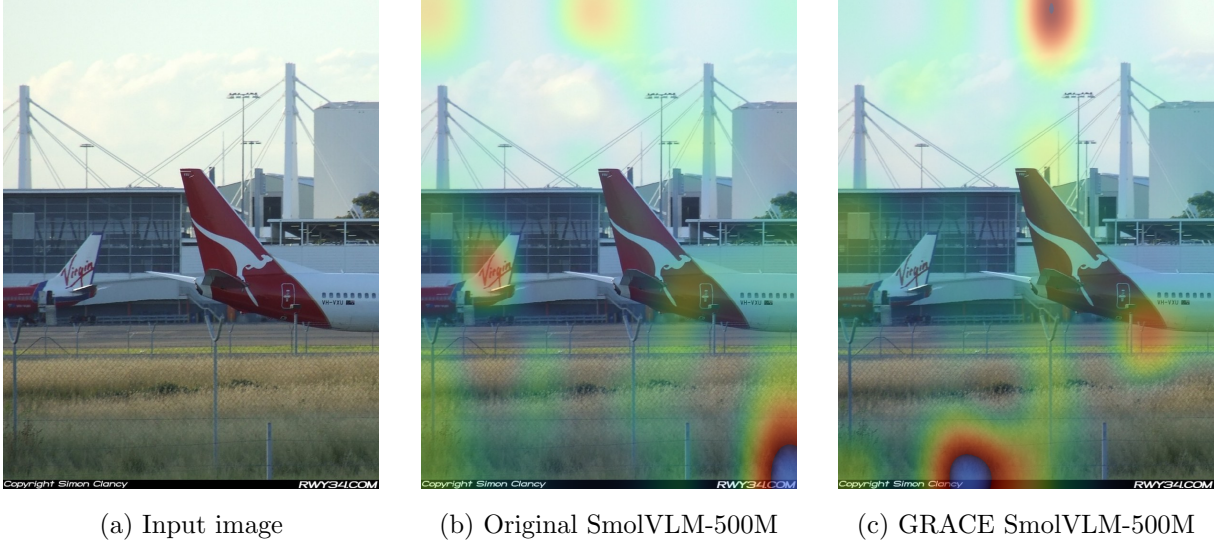


Figure 5.3: Representative TextVQA attention example.

### Answer to RQ1

GRACE improves over direct finetuning, same-family logits KD, ULD and the compared heterogeneous multi-teacher baselines in Table 5.3. The score order places same-family logits baselines below cross-family ULD, cross-family ULD below most multi-teacher settings and GRACE above every compared baseline. Within the multi-teacher group, GRACE scores above SHM-CKA, REINFORCE and Uniform for both student sizes across the three benchmarks. The largest gains come from combining the trie-Wasserstein loss with gradient-based alignment.

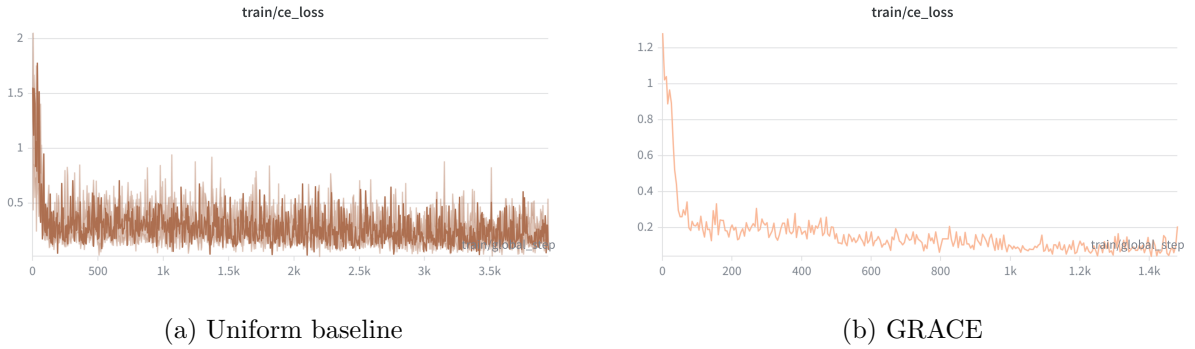
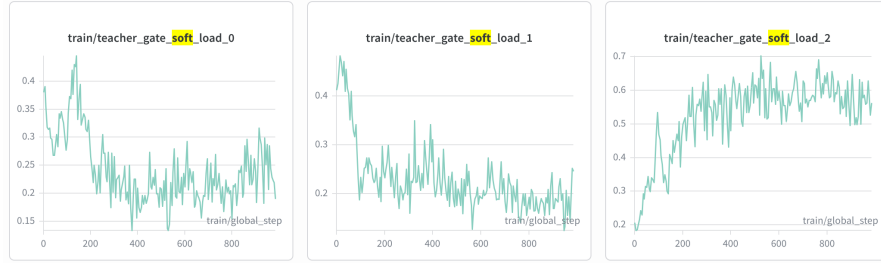


Figure 5.4: Uniform and GRACE training curves.

Figure 5.4 shows that the GRACE run produces a smoother and more stable training curve than Uniform on DocVQA. The Uniform curve shows sharper oscillations, while the GRACE curve decreases more steadily after the early training phase.



(a) Without GRACE refinement



(b) With GRACE

Figure 5.5: Routed-teacher load examples.

Figure 5.5 contrasts collapsed and stable routing behavior. Without the full GRACE recipe, the routed load concentrates on one teacher, reducing teacher diversity. With GRACE, the load stays distributed across the teacher pool, so the student continues to receive heterogeneous supervision. The relative gains in Table 5.4 are consistent with this routing behavior. Relative gain is computed as  $(\text{GRACE} - \text{baseline})/\text{baseline}$ .

Table 5.4: Relative gains of GRACE.

| Benchmark           | vs Original      | vs Finetuning    | vs Uniform      | vs Highest non-GRACE |
|---------------------|------------------|------------------|-----------------|----------------------|
| <i>SmolVLM-500M</i> |                  |                  |                 |                      |
| TextVQA             | +12.63 (+22.26%) | +12.42 (+21.80%) | +5.50 (+8.61%)  | +5.38 (+8.41%)       |
| DocVQA              | +13.59 (+22.83%) | +11.15 (+18.00%) | +3.52 (+5.06%)  | +3.52 (+5.06%)       |
| ChartQA             | +17.40 (+27.71%) | +17.20 (+27.30%) | +4.60 (+6.08%)  | +3.32 (+4.32%)       |
| <i>SmolVLM-256M</i> |                  |                  |                 |                      |
| TextVQA             | +13.56 (+28.75%) | +10.90 (+21.88%) | +6.16 (+11.29%) | +4.42 (+7.85%)       |
| DocVQA              | +11.84 (+24.34%) | +18.45 (+43.90%) | +4.78 (+8.58%)  | +1.95 (+3.33%)       |
| ChartQA             | +27.28 (+60.57%) | +22.64 (+45.57%) | +8.64 (+13.57%) | +3.58 (+5.21%)       |

The gains over the original SmolVLM-500M student are large: +22.26% on TextVQA, +22.83% on DocVQA and +27.71% on ChartQA. The gains over direct finetuning are almost as large, at +21.80%, +18.00% and +27.30%, respectively. The gain over static uniform averaging is smaller, but it remains consistent across all three tasks, isolating the value of adaptive teacher weighting from the value of simply using more than one teacher. The same trend holds for SmolVLM-256M, with relative gains of +28.75%, +24.34% and +60.57% on the three benchmarks. For the gap-to-teacher analysis, let the reference point be the highest-scoring teacher in the GRACE comparison pool reported in Table 5.2. Let  $g_{\text{orig}} = y_T - y_{\text{orig}}$  and  $g_{\text{grace}} = y_T - y_{\text{grace}}$ . The fraction of the gap closed is  $1 - g_{\text{grace}}/g_{\text{orig}}$ .

Table 5.5: Gap-to-teacher analysis for SmolVLM-500M.

| Benchmark | Highest teacher | Original gap | GRACE gap | Gap closed |
|-----------|-----------------|--------------|-----------|------------|
| TextVQA   | 79.70           | 22.95        | 10.32     | 55.03%     |
| DocVQA    | 93.90           | 34.38        | 20.79     | 39.53%     |
| ChartQA   | 94.24           | 31.44        | 14.04     | 55.34%     |

**The gain is not merely an artifact of comparison with lower-scoring baselines.** On two benchmarks, GRACE removes about half of the original teacher-student gap and even on DocVQA, where the teacher ceiling is highest, it still closes almost 40% of the gap. **The distilled student can outperform individual teachers.** On TextVQA, GRACE reaches 69.38 and exceeds Gemma-3-4B-it at 57.80 by +11.58 points (+20.03%). On ChartQA, it exceeds InternVL3-1B at 75.30 by +4.90 points (+6.51%) and Gemma-3-4B-it at 63.60 by +16.60 points (+26.10%). This shows that routed multi-teacher supervision can combine useful signals that are not available from any single teacher alone.

**Reverse KL consistently underperforms forward KL.** This pattern can be explained by the different student-logit gradients. Let  $p_s = \text{softmax}(z_s)$  be the student distribution and let  $p_t$  be the fixed teacher distribution. For forward KL,

$$\mathcal{L}_{FKL} = \text{KL}(p_t \| p_s) = \sum_v p_t(v) \log \frac{p_t(v)}{p_s(v)}, \quad \frac{\partial \mathcal{L}_{FKL}}{\partial z_s(v)} = p_s(v) - p_t(v).$$

Thus, every teacher-supported token directly pulls the student distribution toward the teacher’s soft target [12]. Reverse KL gives

$$\mathcal{L}_{RKL} = \text{KL}(p_s \| p_t) = \sum_v p_s(v) \log \frac{p_s(v)}{p_t(v)}.$$

The student-logit gradient is

$$\frac{\partial \mathcal{L}_{RKL}}{\partial z_s(v)} = p_s(v) \left( \log \frac{p_s(v)}{p_t(v)} - \mathcal{L}_{RKL} \right).$$

The main difference is the factor  $p_s(v)$ . If  $p_t(v) > 0$  but  $p_s(v) \rightarrow 0$ , then

$$\frac{\partial \mathcal{L}_{FKL}}{\partial z_s(v)} \rightarrow -p_t(v), \quad \frac{\partial \mathcal{L}_{RKL}}{\partial z_s(v)} \rightarrow 0.$$

Teacher-supported tokens that the student currently assigns low probability therefore receive lower-magnitude correction under reverse KL. In finite training, this student-weighted gradient can make reverse KL less useful for answer-token distillation, especially when teacher distributions contain useful secondary mass over OCR variants, numeric formats or plausible answer alternatives. We interpret this as an optimization-bias explanation, not as a universal rule, because recent LLM distillation work shows that the mode-seeking interpretation of reverse KL alone does not fully describe KD behavior [56].

**SHM-CKA alone is not sufficient.** Representation similarity is not the same as task-useful output supervision. CKA measures hidden-state geometry [20], but hidden-state similarity does not imply output equivalence:

$$\text{CKA}(H_s, H_t) \approx 1 \not\Rightarrow p_s(\cdot | x) \approx p_t(\cdot | x).$$

A linear-head example makes this precise. Suppose the teacher representation is an orthogonal rotation of the student representation,  $H_t = H_s R$  with  $R^\top R = I$ . Linear CKA gives

$$\text{CKA}(H_s, H_t) = \frac{\|H_t^\top H_s\|_F^2}{\|H_t^\top H_t\|_F \|H_s^\top H_s\|_F} = 1.$$

However, the answer logits can still differ:

$$z_s = H_s W_s, \quad z_t = H_t W_t = H_s R W_t.$$

Unless  $W_s = R W_t$ , the two models can have identical CKA but different logits and different answer-token distributions. This distinction matches the difference between representational similarity and functional similarity discussed in prior work [32]. In heterogeneous VLM distillation, the final answer distribution also depends on the language head, tokenizer, output vocabulary and task-specific teacher behavior. Although SHM-CKA is competitive on ChartQA, it scores below GRACE on TextVQA and DocVQA, where output-level tokenizer-agnostic matching and sample-dependent teacher weighting are more important. The SHM-CKA results motivate a closer look at representation similarity. We use Centered Kernel Alignment (CKA) as an analysis tool to examine whether teacher and student layers exhibit clear correspondence. For centered hidden-state matrices  $\tilde{\mathbf{H}}_s$  and  $\tilde{\mathbf{H}}_t$ , linear CKA is

$$\text{CKA}(\tilde{\mathbf{H}}_s, \tilde{\mathbf{H}}_t) = \frac{\|\tilde{\mathbf{H}}_t^\top \tilde{\mathbf{H}}_s\|_F^2}{\|\tilde{\mathbf{H}}_t^\top \tilde{\mathbf{H}}_t\|_F \|\tilde{\mathbf{H}}_s^\top \tilde{\mathbf{H}}_s\|_F}. \quad (5.1)$$

Values closer to 1 indicate higher representation similarity [20]. CKA is therefore used here as a representation-analysis tool, not as evidence that two models implement the same answer function. A high CKA score indicates similar hidden geometry on the evaluated samples, but it does not guarantee similar answer-token probabilities after the language head. High internal similarity is not sufficient to justify layer-wise distillation, as the figures and tables below confirm.

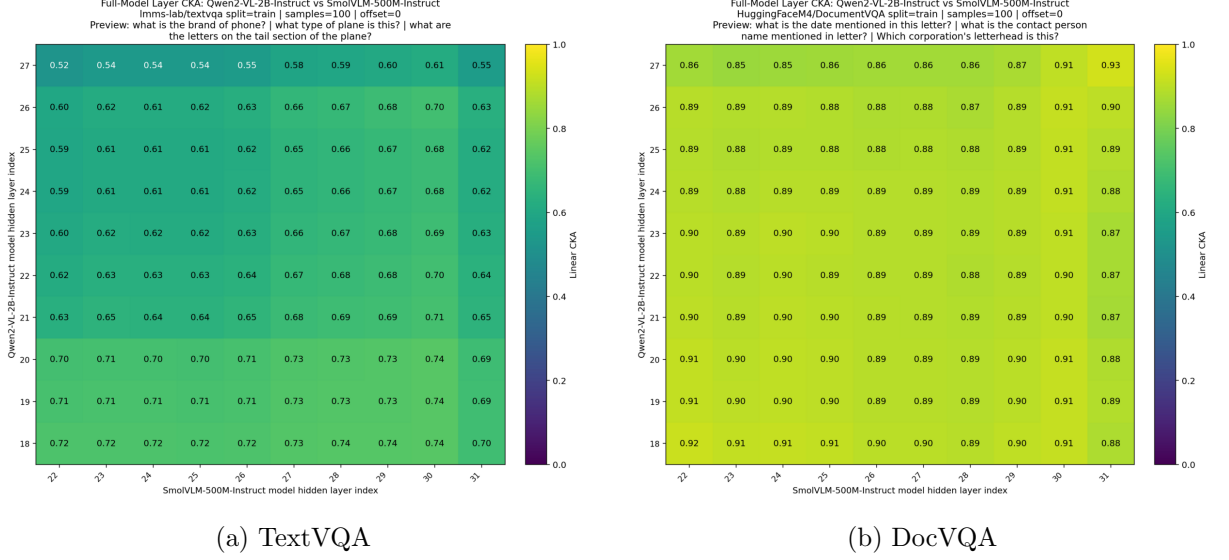
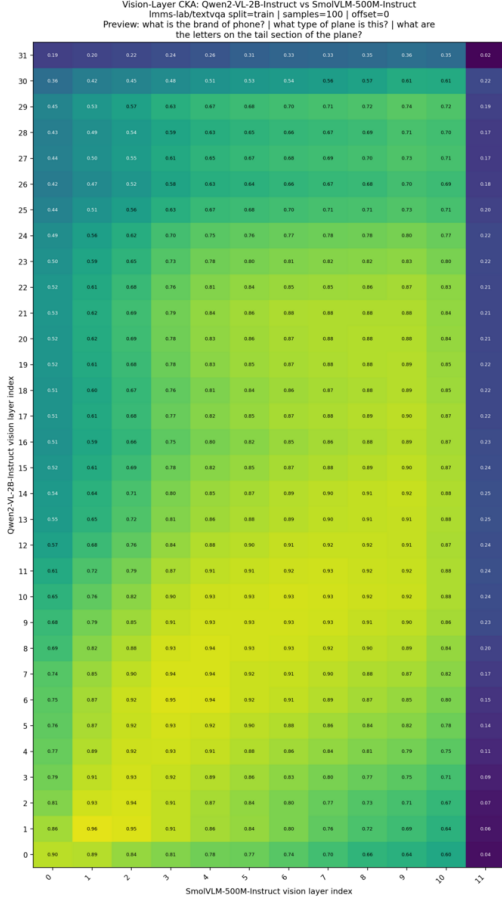
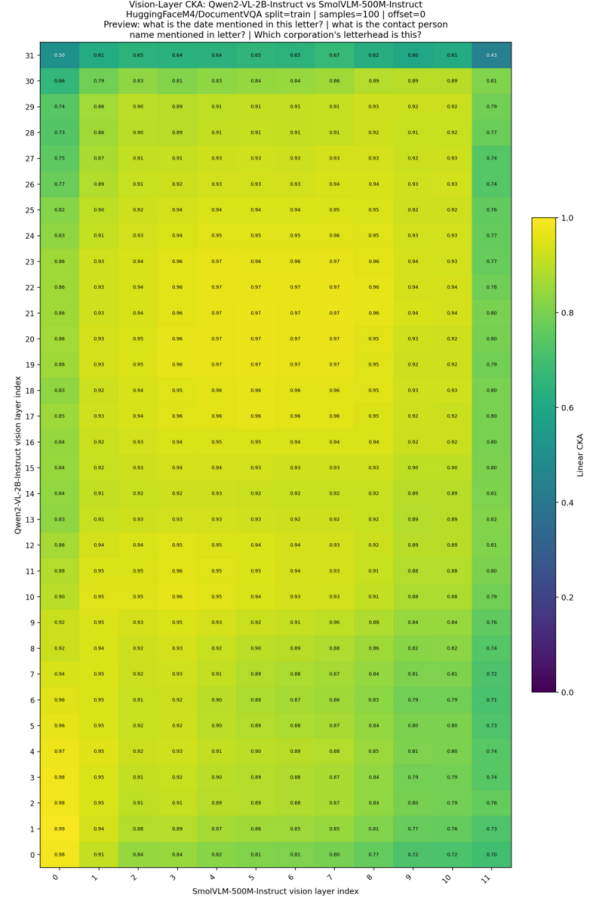


Figure 5.6: Final-layer CKA heatmaps.

Figure 5.6 compares late hidden layers of Qwen2-VL-2B-Instruct and SmolVLM-500M. The TextVQA heatmap contains moderate similarities, mostly around 0.55 to 0.74, with higher values toward later student layers. The DocVQA heatmap is much brighter, with most entries around 0.87 to 0.93. These values show that the final language-side representations are not severely misaligned, especially on document-style inputs.



(a) TextVQA



(b) DocVQA

Figure 5.7: Vision-layer CKA heatmaps.

Figure 5.7 shows the same pattern in the vision stack. Both datasets contain broad high-similarity regions and the DocVQA vision map has higher CKA values across the middle layers. Teacher and student can therefore remain relatively close in both language and vision representations. However, the main benchmark results in Table 5.3 still contain a clear gap between the student models and higher-scoring teachers. The layer-wise baseline also fails to consistently outperform output-level distillation. This pattern supports the design choice in GRACE: output-level tokenizer-agnostic KD and sample-dependent teacher weighting produce higher benchmark scores than representation matching alone, because high hidden-state similarity does not guarantee that direct layer matching will produce better task transfer.

## 5.2 Ablation Study

This section answers RQ2 from Chapter 1 by isolating the main components of GRACE on DocVQA. Section 4.4 defines the ablation variants, with the corresponding results for both student models reported in Table 5.6.

Table 5.6: DocVQA component ablation.

| DocVQA ablation variant                           | SmolVLM      |              |
|---------------------------------------------------|--------------|--------------|
|                                                   | 256M         | 500M         |
| Cross-family ULD (single teacher)                 | 52.24        | 65.28        |
| Uniform averaging                                 | 55.70        | 69.59        |
| Routing only + trie-Wasserstein                   | 57.45        | 63.23        |
| Full GRACE + trie-Wasserstein                     | <b>60.48</b> | <b>73.11</b> |
| w/o EMA ( $\beta_{\text{EMA}} = 0$ )              | 57.45        | 71.25        |
| w/ fixed blend ( $\lambda_{\text{blend}} = 1.0$ ) | 57.97        | 70.45        |

The variants form a cumulative sequence of increasingly specific controls. The single-teacher ULD row tests whether cross-family token distribution transfer is useful before any teacher mixture is introduced. Uniform averaging then tests the value of teacher diversity without any adaptive weighting. The routing-only variant adds sample-conditioned teacher selection, while the full method adds the post-routing gradient check. The two final rows isolate the refinement module itself by removing temporal smoothing or by fixing the blend to rely only on routed weights over the active set. This ordering separates the benefit of using more teachers from the benefit of deciding which teacher signals should actually shape the update. **Teacher diversity is necessary but not sufficient because uniform multi-teacher averaging adds useful supervision while still failing to resolve tokenizer mismatch, teacher specialization and conflicting KD directions.** Across the ablation results in Table 5.6, the full method is the highest-scoring setting for both student sizes. Moving from a single cross-family teacher (ULD) to uniform four-teacher averaging adds +3.46 on SmolVLM-256M and +4.31 on SmolVLM-500M. The gain shows that multiple heterogeneous teachers provide higher-scoring supervision than any single teacher, but uniform weighting leaves a further gap of +4.78/+3.52 to the full method.

**Routing without gradient refinement.** On SmolVLM-500M, routing-only scoring (63.23) is *worse* than uniform averaging (69.59) by  $-6.36$  points. The router successfully identifies candidate teachers, but without gradient refinement it assigns incorrect relative weights and amplifies the contribution of conflicting teachers. Without GRACE, these wrong weights degrade the result below a naive equal average. On SmolVLM-256M the pattern reverses: routing-only (57.45) beats uniform (55.70) by  $+1.75$ . SmolVLM-500M, the larger student, produces a sharper routing distribution that amplifies weighting errors when

uncorrected. SmolVLM-256M produces a flatter routing signal closer to uniform, so routing-only is less damaging and even slightly beneficial. This capacity-dependent behavior shows that a larger student representation does not automatically produce safer teacher weighting. A larger student may give the router a more separable representation of the sample, but the resulting gate can still be confidently wrong when teacher relevance and objective compatibility diverge. The smaller student appears to produce a less peaked routing signal, which behaves closer to a softened uniform mixture. This supports the design choice of treating routing as a proposal step and gradient-agreement refinement as a validation step.

**Gradient-agreement refinement acts as the critical corrective layer because it adjusts routed teacher weights according to whether each teacher’s KD direction agrees with the supervised task objective.** Full GRACE recovers +9.88 points over routing-only on SmolVLM-500M and +3.03 on SmolVLM-256M. Decomposing the 500M gain, the active-set router-weight variant ( $\lambda_{\text{blend}} = 1.0$ ) already recovers +7.22 points (63.23  $\rightarrow$  70.45) by replacing uniform post-routing weights with learned router weights while retaining the GRACE active-set filter. Full score-softmax blending then adds another +2.66 points (70.45  $\rightarrow$  73.11). On SmolVLM-256M, the active-set router-weight variant adds only +0.52 points (57.45  $\rightarrow$  57.97), while full score-based blending adds +2.51 points (57.97  $\rightarrow$  60.48). These results show that active-set filtering and router weighting help, but agreement-score weighting is still needed for the highest final score.

Removing EMA smoothing ( $\beta_{\text{EMA}} = 0$ ) degrades both students (256M: 60.48  $\rightarrow$  57.45, 500M: 73.11  $\rightarrow$  71.25), confirming that temporal smoothing stabilizes score estimates but is secondary to the refinement step itself. The fixed-blend and no-EMA variants separate two roles of the refinement module. The fixed-blend variant keeps the active-set restriction but removes the score-softmax influence, so it tests whether filtering alone is enough. Its lower score shows that selecting an active teacher set is useful but incomplete. The no-EMA variant keeps score weighting but removes temporal smoothing, showing that batch-level gradient agreement is noisy and benefits from accumulated teacher-level history. Together, these two controls indicate that GRACE needs both a teacher-compatibility signal and a stable way to use that signal across training updates.

## Answer to RQ2

The ablation isolates three cumulative contributions: (1) heterogeneous teacher diversity (+3 to +4 pts over single-teacher ULD), (2) sparse routing structure (beneficial only when paired with refinement) and (3) gradient-agreement refinement, which adds +3 to +10 pts over routing-only and is the single largest contributor. EMA smoothing provides additional stabilization by reducing the effect of noisy batch-level gradient-agreement scores before they are converted into final KD weights. Without gradient refinement, routing can degrade performance below uniform averaging.

### 5.3 Efficiency Analysis

This section addresses RQ3 from Chapter 1. Runtime, training cost and storage overhead are analyzed using the H200 rental rate and the observed Cloudflare R2 storage footprint. **Component-level overhead.** GRACE adds online overhead from three sources: sparse trie-Wasserstein support construction, teacher routing and gradient-agreement scoring. The trie-Wasserstein term is the dominant online algorithmic addition because it must compare sparse student and teacher output supports at supervised answer positions. Its cost depends on the retained top- $K_{TW}$  support, the number of active trie edges and the number of routed teachers. The teacher router is comparatively small because it is an MLP over pooled student hidden states, so its cost is minor relative to the student forward and backward pass. Gradient-agreement refinement is more expensive than ordinary fixed weighting because it scores teacher-specific KD directions. This cost scales with the number of teacher-specific KD losses being scored, while top- $k$  routing keeps the final KD aggregation sparse.

**Offline versus online cost.** The implementation separates teacher inference from student training. Teacher logits are generated offline and stored before distillation, so the online training loop does not need to run the frozen teachers during every student update. This reduces online GPU memory pressure and makes the reported experiments feasible on a single high-memory GPU. The tradeoff is that cost moves to the offline cache-generation stage and to storage and I/O. This design is attractive when the same cached teacher logits are reused across multiple ablations, since the expensive teacher forward passes are paid once. It becomes less attractive when datasets, prompts or teacher pools change frequently, because the corresponding cached logits must be regenerated.

**Scalability.** The main scaling pressure comes from the product of dataset size, number of teachers and cached answer-token logits. Adding teachers increases offline generation time, remote storage and the number of candidate KD losses that may need to be scored. Increasing  $K_{TW}$  raises the online cost of sparse trie construction, while increasing

$k_{\text{route}}$  raises the number of teacher losses used per sample. The current four-teacher setting is a workable operating point in this tradeoff: it is large enough to expose teacher specialization, but still small enough for sparse routing, gradient refinement and remote-cache streaming to remain manageable.

### Answer to RQ3

GRACE offloads teacher inference to a one-time offline stage, making online GPU cost comparable to standard finetuning. Per answer position, trie-Wasserstein matching costs  $O(K_{\text{TW}}\bar{L})$  over retained token paths, routing adds  $O(d_h N)$  for the MLP gate over  $N$  teachers, and gradient-agreement scoring adds  $O(N|\Theta|)$  for per-teacher cosine comparisons. Top- $k$  routing bounds the active teacher count per sample at  $k_{\text{route}} \ll N$ , keeping aggregation sparse. Offline logit generation is paid once and amortized across all subsequent runs, offering a favorable cost-performance tradeoff for iterative experimentation.

## 5.4 Discussion

Taken together, the efficiency results and benchmark tables show that across all three benchmarks and both student sizes, the four method families produce a consistent performance ordering. Same-family KL baselines obtain the lowest scores, single cross-family distillation scores higher, uniform multi-teacher weighting improves further and GRACE obtains the highest scores overall [12, 17]. No single component fully addresses the heterogeneous distillation problem: heterogeneous teachers improve scores, but the student still needs a tokenizer-agnostic loss, sample-dependent routing and a stable post-routing weighting scheme in the evaluated setting. The training recipe explains this pattern because GRACE combines a shared-trie Wasserstein loss for mismatched tokenizers with routed multi-teacher selection, top- $k$  masking, routed-weight renormalization and EMA-smoothed gradient-agreement refinement [17, 33, 37]. The smoother DocVQA training curve in Figure 5.4 is consistent with that design: the router narrows the teacher set first, then GRACE reweights only the routed teachers, reducing the score oscillations caused by uniformly mixing heterogeneous teacher signals.

**First-order interpretation of gradient-agreement refinement.** The gradient-agreement component can also be understood from a first-order multi-objective view. Let  $g_{CE} = \nabla_{\theta} \mathcal{L}_{CE}$  and  $g_i = \nabla_{\theta} \mathcal{L}_{KD}^{(i)}$  be the supervised and teacher-specific KD gradients.

A combined update with teacher weights  $w_i$  is

$$\Delta\theta = -\eta \left( g_{CE} + \alpha_{KD} \sum_i w_i g_i \right).$$

The first-order change in the supervised loss is

$$\Delta\mathcal{L}_{CE} \approx g_{CE}^\top \Delta\theta = -\eta \|g_{CE}\|_2^2 - \eta \alpha_{KD} \sum_i w_i \langle g_{CE}, g_i \rangle.$$

If  $\langle g_{CE}, g_i \rangle < 0$ , then teacher  $i$  contributes a term that increases the first-order CE loss when the supervised objective should decrease. This is the gradient-conflict case studied in multi-task optimization [42]. GRACE handles this conflict through teacher weighting with cosine agreement:

$$\cos(g_{CE}, g_i) = \frac{\langle g_{CE}, g_i \rangle}{\|g_{CE}\|_2 \|g_i\|_2}.$$

Teachers with  $\cos(g_{CE}, g_i) < 0$  are suppressed before they distort the supervised update, which explains why gradient refinement improves over routing alone. The largest gain appears on ChartQA and the trend holds for SmolVLM-256M as well. Sample-dependent teacher assignment has the largest observed benefit when the task mixes visual parsing and numerical reasoning. The method also benefits both evaluated student models. The comparison with SHM-CKA reinforces this conclusion by showing that heterogeneous teacher access is not enough when the method relies only on hidden-state matching with no routed output-level KD. SHM-CKA is competitive on ChartQA yet scores lower than GRACE on TextVQA and DocVQA. Together with the CKA heatmaps, this confirms that internal similarity alone does not guarantee effective transfer. **Sample-dependent teacher weighting therefore remains necessary, particularly when the teacher pool contains models with different tokenizers, architectures and task-specific strengths** [20, 31]. The remaining limitations and future directions are discussed in the concluding chapter.

# Conclusion and Future Work

## Summary

This thesis studied knowledge distillation for heterogeneous vision-language models, where teacher utility varies across samples and teacher-student tokenizers do not match. Building on cross-tokenizer distillation, optimal transport and sparse routing [17, 33, 37], the work introduced Gradient-Routed Alignment with Capacity-aware Experts (GRACE), a routed multi-teacher framework that combines three main contributions: trie-Wasserstein tokenizer-agnostic distillation, sparse teacher routing and gradient-agreement refinement. The first contribution is a geometry-aware cross-tokenizer distillation objective. Trie-Wasserstein KD maps decoded teacher and student tokens into a shared byte-trie structure and compares subtree probability masses, so token distributions can be matched across different vocabularies. The second contribution is an objective-aware post-routing refinement mechanism. Sparse routing selects sample-relevant teachers, while gradient-agreement refinement adjusts routed teacher weights according to alignment between each teacher’s KD gradient and the supervised answer gradient. The third contribution is an empirical study showing that output-level tokenizer-agnostic distillation with routing and gradient agreement outperforms the compared same-family, single-teacher and heterogeneous multi-teacher baselines in the evaluated VQA setting.

Across TextVQA, DocVQA and ChartQA, GRACE obtains the highest scores among the compared methods in Chapter 5. For SmolVLM-500M, it reaches 69.38 on TextVQA, 73.11 on DocVQA and 80.20 on ChartQA. For SmolVLM-256M, it reaches 60.73, 60.48 and 72.32 on the same three benchmarks. On SmolVLM-500M, GRACE closes about 55% of the original teacher-student gap on TextVQA and ChartQA and about 40% on DocVQA.

## Limitations

Despite these results, several limitations remain that future work should address. **Cost-limited evaluation** is the first limitation. The reported numbers are single-run results, so the thesis does not provide confidence intervals or repeated-seed estimates for small gaps between high-scoring baselines. As shown in Section 5.3, an average run costs approximately 14.41 USD before offline cache generation and storage cost. Although the main tables use a shared four-teacher pool and consistent hyperparameters, the absence of repeated seeds limits strict uncertainty claims about closely matched methods.

**Scope of validation** is also limited. The empirical study focuses on VQA benchmarks and SmolVLM-family students. The results support text-rich VQA distillation with compact

projector-based students, but they do not establish whether the same behavior holds for other multimodal task families, other student architectures or larger teacher pools. Tasks with long-form answers, multi-image inputs, video reasoning or open-ended dialogue may place different demands on teacher selection and answer-token supervision.

**Reasoning-oriented models are not explicitly supported.** GRACE currently distills teacher supervision through answer-token output distributions and gradient agreement with the supervised answer loss. This design is appropriate for the evaluated VQA setting, where the target is a short final answer. It does not yet explicitly support reasoning-oriented or “thinking” models whose behavior depends on intermediate deliberation traces, hidden reasoning tokens or extended test-time reasoning. If a teacher produces only a final answer distribution, GRACE can still use that distribution as a teacher signal. However, the method does not distinguish between reasoning-phase tokens and answer-phase tokens, does not route teachers based on the quality of intermediate reasoning and does not define a distillation objective for multi-step thought trajectories.

## Future Work

Future work should extend the thesis in four directions. First, stronger evaluation should use repeated runs with multiple random seeds, report mean and standard deviation and use paired comparisons when the same cached teacher logits are reused. This would make the smaller differences between close multi-teacher baselines easier to interpret.

Second, systems work should reduce the storage, I/O and gradient-computation costs of heterogeneous teacher distillation. More compact logit storage, adaptive cache loading and lower-cost agreement scoring would make the method easier to scale to more datasets and larger teacher pools.

Third, future method extensions should move beyond answer-token-centric distillation. Visual-token alignment, cross-attention transfer and region-level matching could complement output-level KD, especially for tasks where the student must localize evidence before producing a short answer.

Fourth, support for reasoning-oriented models should be studied with process-level distillation mechanisms. Such extensions would need to separate reasoning-stage information from final-answer generation and define how teachers are routed or weighted when intermediate reasoning traces are available. Future work can also explore richer router inputs, improved objective-conflict handling and teacher pools that mix specialized document, chart, general VLM and reasoning-oriented experts [21–23].

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# Appendix A

## Proof of the Closed-Form Trie-Wasserstein

This appendix derives the closed-form expression used by the KD loss in Chapter 3. The result is stated for a general rooted tree [18, 33].

The proof uses the following notation. The tree  $T = (V, E, r)$  is a rooted tree with root  $r$  and each edge  $e \in E$  has weight  $w(e) > 0$ . For an edge  $e = (p(e), c(e))$ , define  $\Gamma(e) = \{v \in V \mid e \text{ lies on the path from } r \text{ to } v\}$ , the subtree below  $e$ . For vertices  $u, v \in V$ , the tree metric is  $d_T(u, v) = \sum_{e \in \text{path}(u, v)} w(e)$ .

$\mu$  and  $\nu$  are probability measures on  $V$ . In the trie-Wasserstein setting, these are the student and teacher token distributions on the leaves of the shared byte trie, with zero mass assigned to non-leaf vertices. The Kantorovich form of the  $W_1$  distance is

$$W_{1,T}(\mu, \nu) = \min_{\pi \in \Pi(\mu, \nu)} \sum_{u, v \in V} \pi(u, v) d_T(u, v),$$

where  $\Pi(\mu, \nu)$  is the set of couplings satisfying

$$\sum_{v \in V} \pi(u, v) = \mu(u), \quad \sum_{u \in V} \pi(u, v) = \nu(v), \quad \pi(u, v) \geq 0.$$

For each edge  $e$ , the subtree imbalance is  $\Delta(e) = \mu(\Gamma(e)) - \nu(\Gamma(e))$ . We prove that

$$W_{1,T}(\mu, \nu) = \sum_{e \in E} w(e) |\Delta(e)|.$$

We fix any feasible coupling  $\pi \in \Pi(\mu, \nu)$  and any edge  $e \in E$ . Removing  $e$  separates the tree into two components:  $\Gamma(e)$  and  $V \setminus \Gamma(e)$ . Any mass transported from one side of this cut to the other must cross  $e$ .

The outward and inward crossing masses are

$$q_\pi^{\text{out}}(e) = \sum_{u \in \Gamma(e)} \sum_{v \notin \Gamma(e)} \pi(u, v), \quad q_\pi^{\text{in}}(e) = \sum_{u \notin \Gamma(e)} \sum_{v \in \Gamma(e)} \pi(u, v).$$

Thus the total mass crossing  $e$  is  $q_\pi(e) = q_\pi^{\text{out}}(e) + q_\pi^{\text{in}}(e)$ .

The marginal constraints imply that the source mass in  $\Gamma(e)$  is

$$\mu(\Gamma(e)) = \sum_{u \in \Gamma(e)} \sum_{v \in \Gamma(e)} \pi(u, v) + q_\pi^{\text{out}}(e),$$

The target mass in  $\Gamma(e)$  has the analogous decomposition

$$\nu(\Gamma(e)) = \sum_{u \in \Gamma(e)} \sum_{v \in \Gamma(e)} \pi(u, v) + q_\pi^{\text{in}}(e).$$

Subtracting gives  $\Delta(e) = q_\pi^{\text{out}}(e) - q_\pi^{\text{in}}(e)$ . Therefore,  $|\Delta(e)| \leq q_\pi^{\text{out}}(e) + q_\pi^{\text{in}}(e) = q_\pi(e)$ . So every feasible transport plan must move at least  $|\Delta(e)|$  mass across edge  $e$ .

Since the path between any two vertices in a tree is unique,  $d_T(u, v) = \sum_{e \in E} w(e) \mathbf{1}\{e \in \text{path}(u, v)\}$ . The cost of  $\pi$  therefore decomposes as

$$\sum_{u, v \in V} \pi(u, v) d_T(u, v) = \sum_{e \in E} w(e) q_\pi(e).$$

The inequality  $q_\pi(e) \geq |\Delta(e)|$  gives

$$\sum_{u, v \in V} \pi(u, v) d_T(u, v) \geq \sum_{e \in E} w(e) |\Delta(e)|.$$

Minimizing over all feasible  $\pi$  gives

$$W_{1,T}(\mu, \nu) \geq \sum_{e \in E} w(e) |\Delta(e)|.$$

It remains to construct a feasible plan attaining this lower bound. The signed measure is  $\sigma(v) = \mu(v) - \nu(v)$ . Since  $\mu$  and  $\nu$  are probability measures,  $\sum_{v \in V} \sigma(v) = 0$ . For each edge  $e$ ,  $\Delta(e) = \sum_{v \in \Gamma(e)} \sigma(v)$ . Thus  $\Delta(e)$  is the net surplus mass in the subtree below  $e$ . The quantity  $\Delta(e)$  can be interpreted as a signed flow across edge  $e$ . If  $\Delta(e) > 0$ , then  $\Delta(e)$  units of surplus mass move upward from  $\Gamma(e)$  toward the root. If  $\Delta(e) < 0$ , then  $|\Delta(e)|$  units of mass move downward into  $\Gamma(e)$ . Because the total signed mass is zero, the total surplus equals the total deficit, so this flow can satisfy all demands. More explicitly, for each non-root vertex  $x$ , the edge to its parent is  $e_x = (p(x), x)$ , and the flow is  $F(e_x) = \Delta(e_x)$ . The subtree below  $e_x$  consists of  $x$  and the subtrees below the child edges of  $x$ . Therefore,

$$F(e_x) = \sigma(x) + \sum_{y \in \text{Ch}(x)} F(e_y).$$

This identity expresses flow conservation: the net mass sent through the parent edge of  $x$  equals the local surplus at  $x$  plus the surplus received from its children. At the root, the same conservation condition follows from  $\sum_{v \in V} \sigma(v) = 0$ .

The signed edge flow  $F$  transports the positive part of  $\sigma$  to the negative part of  $\sigma$ . Since  $T$  is a tree, this edge flow can be decomposed into transport paths from surplus vertices to deficit vertices.  $\pi^*$  denotes the corresponding coupling. This coupling leaves common mass fixed at each vertex and the fixed mass has zero cost. At vertex  $v$ , this mass equals  $\min\{\mu(v), \nu(v)\}$ , so  $\pi^* \in \Pi(\mu, \nu)$ . By construction, the amount of mass transported across each edge  $e$  under  $\pi^*$  is exactly  $|\Delta(e)|$ . Therefore,

$$\sum_{u, v \in V} \pi^*(u, v) d_T(u, v) = \sum_{e \in E} w(e) |\Delta(e)|.$$

This construction gives the upper bound

$$W_{1,T}(\mu, \nu) \leq \sum_{e \in E} w(e) |\Delta(e)|.$$

Combining the lower and upper bounds yields

$$W_{1,T}(\mu, \nu) = \sum_{e \in E} w(e) |\mu(\Gamma(e)) - \nu(\Gamma(e))|.$$

# Appendix B

## Implementation Details

This appendix lists the optimization and routing hyperparameters used across all distillation experiments and records the engineering details needed to reproduce the training infrastructure. The hyperparameters correspond to the evaluation protocol described in Chapter 4.

Table B.1: Experiment parameters.

| Setting                            | Value                                                                                                |
|------------------------------------|------------------------------------------------------------------------------------------------------|
| <i>Model setup</i>                 |                                                                                                      |
| Student model                      | HuggingFaceTB/SmolVLM-500M-Instruct                                                                  |
| Teacher models                     | Qwen/Qwen2.5-VL-3B-Instruct, Qwen/Qwen2-VL-2B-Instruct, OpenGVLab/InternVL3-1B, google/gemma-3-4b-it |
| Teacher weighting strategy         | routing (GRACE path)                                                                                 |
| Distillation loss                  | Trie-Wasserstein                                                                                     |
| Trie-Wasserstein rho               | 0.7                                                                                                  |
| Trie-Wasserstein top- $k$          | 64                                                                                                   |
| Validation split                   | validation                                                                                           |
| <i>Hardware and infrastructure</i> |                                                                                                      |
| Hardware and runtime               | Single-GPU H200 or Vast.ai runs                                                                      |
| Distributed training               | DeepSpeed ZeRO-2                                                                                     |
| Numerical precision                | bf16 enabled, half precision (fp16) disabled, TF32 enabled                                           |
| FlashAttention-2                   | Enabled                                                                                              |
| Training data format               | LLaVA-style multimodal instruction JSON                                                              |
| <i>Training</i>                    |                                                                                                      |
| Max sequence length                | 16384                                                                                                |
| Training epochs                    | 2                                                                                                    |
| Per-device batch size              | 80                                                                                                   |
| Gradient accumulation steps        | 1                                                                                                    |
| Effective batch size               | 80                                                                                                   |
| Gradient checkpointing             | Enabled                                                                                              |
| Frozen modules                     | None                                                                                                 |
| Dataloader workers                 | 4                                                                                                    |
| Logging and checkpointing          | Log every 5 steps and save every 150 steps                                                           |

*Continued on the next page.*

Table B.1 continued from the previous page.

| Setting                         | Value                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------|
| Seed                            | 42                                                                                                    |
| <i>Optimizer</i>                |                                                                                                       |
| Optimizer                       | adamw_torch                                                                                           |
| Adam betas                      | (0.9, 0.98)                                                                                           |
| Adam epsilon                    | $1 \times 10^{-7}$                                                                                    |
| Weight decay                    | 0.0                                                                                                   |
| Base learning rate              | $1 \times 10^{-5}$                                                                                    |
| Vision learning rate            | $2 \times 10^{-6}$                                                                                    |
| Connector learning rate         | $1 \times 10^{-5}$                                                                                    |
| Learning-rate schedule          | Cosine decay with warmup ratio 0.03                                                                   |
| <i>GRACE hyperparameters</i>    |                                                                                                       |
| Student/teacher temperature     | $\tau_s = \tau_t = 1.0$                                                                               |
| KD weight                       | $\alpha_{\text{KD}} = 0.5$                                                                            |
| Top- $k$ routing                | $k_{\text{route}} = 2$ , with top-scoring teacher masking and selected softmax-weight renormalization |
| Gate entropy weight             | $\lambda_{\text{ent}} = 10^{-3}$                                                                      |
| Router $z$ -loss weight         | $\lambda_z = 10^{-3}$                                                                                 |
| GRACE warmup ratio              | $\rho_g = 0.05$                                                                                       |
| GRACE softmax beta              | $\beta_{\text{score}} = 4.0$                                                                          |
| GRACE blend exponent            | $\lambda_{\text{blend}} = 0.9$                                                                        |
| GRACE EMA decay                 | $\beta_{\text{EMA}} = 0.8$                                                                            |
| Score-spread fallback threshold | $\epsilon_{\text{fb}} = 0.01$                                                                         |
| Alignment threshold             | $\delta = 0.15$                                                                                       |

## B.1 Baseline Implementation

This section records the additional baseline definitions used for the comparison methods in Chapter 4, including how their teacher selection, weighting and auxiliary loss terms are implemented for fair comparison.

**SHM-CKA.** The baseline first loads an offline CKA matrix and, for each student layer  $s$ , selects the top- $k$  teacher layers  $\mathcal{T}_s$ . Their CKA scores are normalized into soft weights  $\alpha_{s,t}$ . The per-teacher layer loss is

$$\mathcal{L}_{\text{layer}}^{(i)} = \frac{1}{|S|} \sum_{s \in S} \sum_{t \in \mathcal{T}_s} \alpha_{s,t} (1 - \sqrt{\text{CKA}(\mathbf{H}_s, \mathbf{H}_t)}). \quad (\text{B.1})$$

SHM-CKA also assigns each teacher a ULD logits loss. The averaged terms are

$$\bar{\mathcal{L}}_{\text{ULD}} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{\text{ULD}}^{(i)}, \quad \bar{\mathcal{L}}_{\text{layer}} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{\text{layer}}^{(i)}. \quad (\text{B.2})$$

The final SHM-CKA objective is

$$\mathcal{L}_{\text{SHM-CKA}} = \frac{1}{3} (\mathcal{L}_{CE} + \bar{\mathcal{L}}_{\text{ULD}} + \bar{\mathcal{L}}_{\text{layer}}). \quad (\text{B.3})$$

**REINFORCE.** The policy uses the pooled student hidden state together with per-teacher summary features. A linear policy predicts independent Bernoulli probabilities

$$\boldsymbol{\pi} = \sigma(W \text{LN}(\mathbf{z}) + b), \quad (\text{B.4})$$

then samples a teacher mask  $m_i \sim \text{Bernoulli}(\pi_i)$ . If no teacher is selected, the highest-probability teacher is used as fallback. The selected mask is normalized into teacher weights

$$w_i = \frac{m_i}{\sum_{j=1}^M m_j}, \quad \mathcal{L}_{KD} = \sum_{i=1}^M w_i \mathcal{L}_{KD}^{(i)}. \quad (\text{B.5})$$

The selector is trained with the policy-gradient term

$$\mathcal{L}_{\text{pol}} = -(r - b) \log p(\mathbf{m} \mid \boldsymbol{\pi}), \quad (\text{B.6})$$

where  $r$  is the reward and  $b$  is an EMA reward baseline. The total reinforced-selection objective is

$$\mathcal{L}_{\text{RS}} = \mathcal{L}_{CE} + \gamma \mathcal{L}_{KD} + \lambda_{\pi} \mathcal{L}_{\text{pol}}. \quad (\text{B.7})$$

## B.2 Offline Teacher-Logit Preparation

This section records the offline teacher-logit generation step used before distillation training. An average full H200 student-training run takes about 7 hours and costs about 14.41 USD at 2.059 USD/hour on Vast.ai, excluding the separate cost of offline teacher-logit generation and remote storage.

Teacher logits are generated *offline* before distillation. An offline caching script runs each frozen teacher on every training example, keeps only the supervised answer-token positions and writes one `.pt` file per sample under a teacher-specific directory. Each record stores the sample index, the selected teacher logits and the corresponding supervised labels. This keeps the online training loop simple: the student only reads precomputed teacher targets, so the teachers do not run again. After generation, the teacher-logit directories are uploaded to a **Cloudflare R2** bucket using its S3-compatible object-storage interface

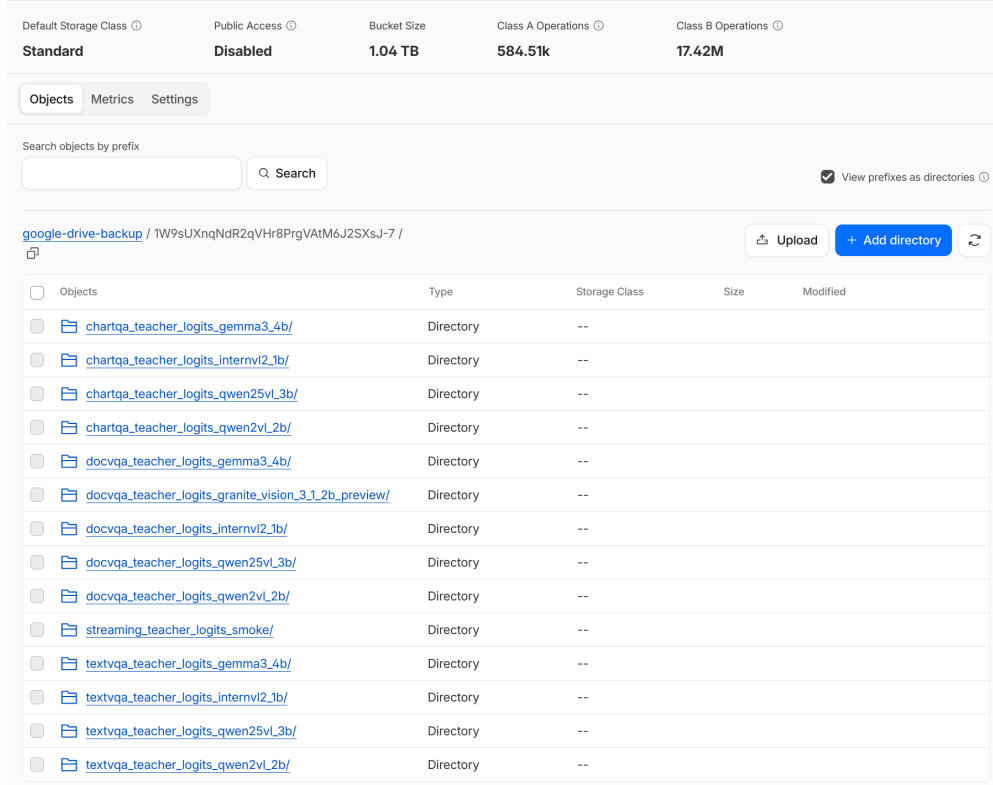


Figure B.1: R2 teacher-logit storage.

[57, 58]. In the active setup, the bucket already exceeds **1 TB**, so copying the full cache to an external GPU instance is not feasible. This is especially relevant on services such as Vast.ai, where local NVMe space is usually much smaller than the full teacher logit store. Figure B.1 shows this offline storage layout and highlights why the training path must stream teacher logits without staging the full cache locally.

### B.3 Remote Logit Streaming During Training

The main training path does *not* download the full cache before training. Figure B.2 shows the full remote-cache path: teacher logits are generated offline and uploaded once, each training step requests a sample, the local cache serves hits directly and cache misses trigger an R2 download followed by least-recently-used eviction when the 100 GB limit is exceeded. When the launcher receives an R2 root in `distill_multi_teachers.sh`, the trainer switches to remote lazy loading. During training, `StreamingTeacherLogitsCache` resolves the teacher-specific remote object path for each requested sample, then fetches files lazily from R2 through `s3fs` and `WholeFileCacheFileSystem` [59, 60]. To keep disk usage bounded, the loader maintains a **100 GB** local cache. Each file is cached on first access, its access time is updated on each subsequent open and the oldest files are evicted once the cache exceeds the limit. The local cache therefore behaves as a small least-recently-used

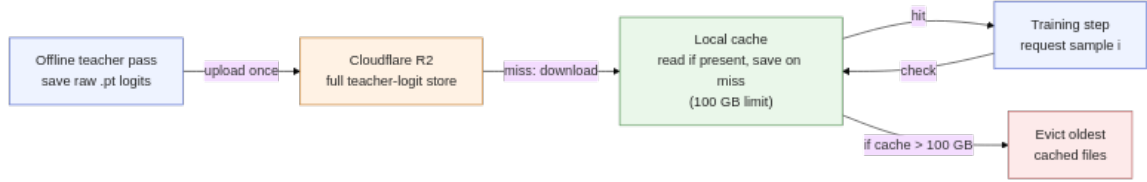


Figure B.2: Training with remote teacher logits.

buffer over the much larger R2 store. A per-teacher file lock guards cache metadata updates so that concurrent dataloader workers do not corrupt the cache state.

Table B.2: Logit-streaming settings.

| Setting               | Value                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------|
| Teacher-logit storage | Cloudflare R2 object storage                                                                |
| Teacher-logit format  | Raw per-sample .pt files                                                                    |
| Remote path template  | remote_root/<br><dataset>_teacher_logits_<br><alias>/<teacher-slug>/<br><index>.pt          |
| Local cache directory | /tmp/teacher-logits-<br>streaming for R2 reads<br>/workspace/cache<br>for local-cache reads |
| Local cache limit     | 100 GB                                                                                      |